

A blurred background image showing a financial chart with a red line graph trending upwards and a yellow dashed line. The chart is overlaid on a dark blue background with some out-of-focus lights.

MODULE 4 : Market-Based Valuation: Price and Enterprise Value Multiples

LEARNING OUTCOMES

- 4.a.** Contrast the method of comparables and the method based on forecasted fundamentals as approaches to using price multiples in valuation and explain economic rationales for each approach.
- 4.b.** Calculate and interpret a justified price multiple.
- 4.c.** Describe rationales for and possible drawbacks to using alternative price multiples and dividend yield in valuation.
- 4.d.** Calculate and interpret alternative price multiples and dividend yield.
- 4.e.** Calculate and interpret underlying earnings, explain methods of normalizing earnings per share (EPS), and calculate normalized EPS.
- 4.f.** Explain and justify the use of earnings yield (E/P).
- 4.g.** Describe fundamental factors that influence alternative price multiples and dividend yield.
- 4.h.** Calculate and interpret a predicted P/E, given a cross-sectional regression on fundamentals, and explain limitations to the cross-sectional regression methodology.

LEARNING OUTCOMES

- 4.i.** calculate and interpret the justified price-to-earnings ratio (P/E), price-to-book ratio (P/B), and price-to-sales ratio (P/S) for a stock, based on forecasted fundamentals.
- 4.j.** calculate and interpret the P/E-to-growth (PEG) ratio and explain its use in relative valuation.
- 4.k.** calculate and explain the use of price multiples in determining terminal value in a multistage discounted cash flow (DCF) model.
- 4.l.** evaluate whether a stock is overvalued, fairly valued, or undervalued based on comparisons of multiples.
- 4.m.** evaluate a stock by the method of comparables and explain the importance of fundamentals in using the method of comparables.
- 4.n.** explain alternative definitions of cash flow used in price and enterprise value (EV) multiples and describe limitations of each definition.
- 4.o.** calculate and interpret EV multiples and evaluate the use of EV/EBITDA.

LEARNING OUTCOMES

- 4.p.** explain sources of differences in cross-border valuation comparisons.
- 4.q.** describe momentum indicators and their use in valuation.
- 4.r.** explain the use of the arithmetic mean, the harmonic mean, the weighted harmonic mean, and the median to describe the central tendency of a group of multiples.

[LOS 4.a] Contrast the method of comparables and the method based on forecasted fundamentals as approaches to using price multiples in valuation and explain economic rationales for each approach.

1.

Method of comparables



Multiple of peer companies ► **Benchmark value of multiples (P/B, P/E,...)**

1 Compare to actual multiple

Actual multiple < Benchmark multiple ► **undervalued**

Actual multiple > Benchmark multiple ► **overvalued**

Actual multiple = Benchmark multiple ► **fairly priced**

2 Multiply by the estimated value of the denominator (Book value, EPS, ...)

The **intrinsic value** of the company

[LOS 4.a] Contrast the method of comparables and the method based on forecasted fundamentals

1. Method of comparables

Example: Method of comparables



Median P/B of peer companies = 2.2 ► **Benchmark B/P = 2.2**

1 Compare to actual multiple

Actual multiple = $2.0 < 2.2$ ► **undervalued**

Actual multiple = $3.0 > 2.2$ ► **overvalued**

Actual multiple = 2.2 ► **fairly priced**

2 Multiply by the estimated value of the denominator BVPS = \$23

The **intrinsic value** of the company = $\$23 \times 2.2 = 50.6$

[LOS 4.a] Contrast the method of comparables and the method based on forecasted fundamentals

2.

Method based on forecasted fundamentals

- Dividend discount model
- Free cash flow valuation model
- Residual income valuation model

Valuation models

Forecasted value (V_0)

Multiple based on valuation model

Intrinsic value (V_0)

Fundamental (Book value, EPS, ..)

compared to

Actual multiple

Market price

Fundamental (Book value, EPS, ..)

Multiple based on valuation model > actual multiple ➤ **undervalued**

Multiple based on valuation model < actual multiple ➤ **overvalued**

Multiple based on valuation model = actual multiple ➤ **fairly priced**

[LOS 4.a] Contrast the method of comparables and the method based on forecasted fundamentals

2.

Method based on forecasted fundamentals

Example: Method based on forecasted fundamentals

- Residual income valuation model

Valuation models

Forecasted value (V_0) = \$46

Multiple based on valuation model

$$\frac{\text{Intrinsic value } (V_0)}{\text{Book value per share}} = \frac{46}{23} = 2.0$$



Actual multiple

$$\frac{\text{Market price}}{\text{Book value per share}}$$

Actual multiple < 2.0



Undervalued

Actual multiple > 2.0



Overvalued

Actual multiple = 2.0



Fairly priced

[LOS 4.b.c.d] Calculate and determine the pros and cons of P/E, P/B, P/S, P/CF, D/P ratios

1.

Price/earning (P/E) ratio

There are two versions defined of P/E ratio: Trailing and Leading P/E with **different denominator**.

Trailing P/E

$$\text{trailing P/E} = \frac{\text{market price per share}}{\text{EPS over previous 12 months}}$$

Leading P/E

$$\text{leading P/E} = \frac{\text{market price per share}}{\text{forecasted EPS over next 12 months}}$$

Advantages and disadvantages

Advantages

- Earnings power is the primary determinant of investment value.
- P/E ratio is popular.
- Empirical research shows that P/E differences are significantly related to long-run stock returns.

Disadvantages

- Earning can be *negative*, which produces meaningless P/E ratio.
- Earning can be volatile, transitory
- Earnings can be distorted, lessen the comparability across firms.

[LOS 4.b.c.d] Calculate and determine the pros and cons of P/E, P/B, P/S, P/CF, D/P ratios

1.

Price/earning (P/E) ratio

Example: Calculating P/E ratio

Byron Investments, Inc., reported €32 million in earnings during the current fiscal year. An analyst forecasts an EPS over the next 12 months of €1.00. Byron has 40 million shares outstanding at a market price of €18.00 per share. Calculate Byron's trailing and leading P/E ratios.

Answer:

$$\text{current-year EPS} = \frac{\text{€}32,000,000}{40,000,000} = \text{€}0.80$$

$$\text{trailing P/E} = \frac{\text{€}18.00}{\text{€}0.80} = 22.5$$

$$\text{leading P/E} = \frac{\text{€}18.00}{\text{€}1.00} = 18.0$$

[LOS 4.b.c.d] Calculate and determine the pros and cons of P/E, P/B, P/S, P/CF, D/P ratios

2.

Price-to-book value (P/B) ratio

Price-to-book
ratio

$$P/B = \frac{\text{market value of equity}}{\text{book value of equity}} = \frac{\text{market price per share}}{\text{book value per share}}$$

Book value of equity = common shareholders' equity
= (total asset – total liabilities)
– preferred stock

Advantages and disadvantages

Advantages

- Book value is usually **positive**.
- Book value is **more stable** than EPS, so P/B can be more useful than P/E when EPS is volatile.
- Book value is an appropriate measure of net asset value for firms that primarily hold **liquid assets**.
- P/B can be useful in valuing companies that are expected to **go out of business**.
- Help explain differences in long-run average stock return.

Disadvantages

- P/Bs **do not reflect** the value of **intangible assets**.
- A firm's business model dictates the size of its asset base. So P/Bs can be misleading.
- Book value is **affected by accounting standards**, reduce the comparability across firms and countries.
- **Inflation and technological change** can cause the book and market values of assets to differ significantly.

[LOS 4.b.c.d] Calculate and determine the pros and cons of P/E, P/B, P/S, P/CF, D/P ratios

2.

Price-to-book value (P/B) ratio

Note that: We often make adjustments to book value to create more useful comparisons of P/B ratios across different stocks, which includes:

- Use tangible book value, which is equal to book value of equity less intangible assets.
- Adjust for significant off-balance-sheet assets and liabilities
- Adjust for differences between the fair and recorded value of assets and liabilities.
- Adjust for accounting differences, for example, between companies using LIFO and FIFO for inventory recognition.

[LOS 4.b.c.d] Calculate and determine the pros and cons of P/E, P/B, P/S, P/CF, D/P ratios

2.

Price-to-book value (P/B) ratio

Example: Calculating P/B ratio

Based on the information in the following figure, calculate the P/B for Crisco Systems, Inc., and Soothsayer Corp. as of the end of 2023.

Data for Crisco Systems, Inc., and Soothsayer Corp.

Company	Book value of equity 2023 (millions)	Sales 2023 (millions)	Shares outstanding 2023 (millions)	Price FYE 2023
<i>Crisco Systems, Inc</i>	\$28,039	\$18,878	7,001	\$17.83
<i>Soothsayer Corp</i>	\$6,320	\$9,475	5,233	\$12.15

[LOS 4.b.c.d] Calculate and determine the pros and cons of P/E, P/B, P/S, P/CF, D/P ratios

2.

Price-to-book value (P/B) ratio

Example: Calculating P/B ratio

Answer:

Crisco Systems, Inc.:

$$\text{Book value per share} = \frac{\text{book value of equity}}{\text{number of shares outstanding}} = \frac{\$28,039}{7,001} = \$4.01$$

$$\text{P/B} = \frac{\text{market price per share}}{\text{book value per share}} = \frac{\$17.83}{\$4.01} = 4.45 \text{ times}$$

Soothsayer Corp.:

$$\text{Book value per share} = \frac{\text{book value of equity}}{\text{number of shares outstanding}} = \frac{\$6,320}{5,233} = \$1.21$$

$$\text{P/B} = \frac{\text{market price per share}}{\text{book value per share}} = \frac{\$12.15}{\$1.21} = 10.04 \text{ times}$$

[LOS 4.b.c.d] Calculate and determine the pros and cons of P/E, P/B, P/S, P/CF, D/P ratios

3.

Price-to-sales (P/S) ratio

Price-to-sales ratio

$$P/S = \frac{\text{market value of equity}}{\text{total sales}} = \frac{\text{market price per share}}{\text{sales per share}}$$

Advantages and disadvantages

Advantages

- P/S is meaningful even for distressed firms. This is not the case for P/E and P/B ratios, which can be negative.
- Sales revenue is not as easy to manipulate or distort as EPS and book value.
- P/S ratios are not as volatile as P/E multiples.

Disadvantages

- High growth in sales does not necessarily indicate high operating profits as measured by earnings and cash flow.
- P/S ratios do not capture differences in cost structures across companies.

[LOS 4.b.c.d] Calculate and determine the pros and cons of P/E, P/B, P/S, P/CF, D/P ratios

3.

Price-to-sales (P/S) ratio

Advantages and disadvantages (continued)

Advantages

- P/S ratios are particularly appropriate for valuing stocks in mature or cyclical industries and start-up companies with no record of earnings. It is also often used to value investment management companies and partnerships.
- Like P/E and P/B ratios, differences in P/S are significantly related to differences in long-run average stock returns.

Disadvantages

- While less subject to distortion, revenue recognition practices can still distort sales forecasts.
For example, analysts should look for company practices that speed up revenue recognition.

[LOS 4.b.c.d] Calculate and determine the pros and cons of P/E, P/B, P/S, P/CF, D/P ratios

3.

Price-to-sales (P/S) ratio

Example: Calculating P/S ratio

Based on the information in the following figure, calculate the current P/S for Crisco Systems, Inc., and Soothsayer Corp.

Data for Crisco Systems, Inc., and Soothsayer Corp.

Company	Book value of equity 2023 (millions)	Sales 2023 (millions)	Shares outstanding 2023 (millions)	Price FYE 2023
<i>Crisco Systems, Inc</i>	\$28,039	\$18,878	7,001	\$17.83
<i>Soothsayer Corp</i>	\$6,320	\$9,475	5,233	\$12.15

[LOS 4.b.c.d] Calculate and determine the pros and cons of P/E, P/B, P/S, P/CF, D/P ratios

3.

Price-to-sales (P/S) ratio

Example: Calculating P/S ratio

Answer:

Crisco Systems, Inc.:

$$\text{Sales per share} = \frac{\text{sales}}{\text{number of shares outstanding}} = \frac{\$18,878}{7,001} = \$2.70$$

$$\text{P/S} = \frac{\text{market price per share}}{\text{sales per share}} = \frac{\$17.83}{\$2.70} = 6.60 \text{ times}$$

Soothsayer Corp.:

$$\text{Book value per share} = \frac{\text{sales}}{\text{number of shares outstanding}} = \frac{\$9,475}{5,233} = \$1.81$$

$$\text{P/S} = \frac{\text{market price per share}}{\text{sales per share}} = \frac{\$12.15}{\$1.81} = 6.71 \text{ times}$$

[LOS 4.b.c.d] Calculate and determine the pros and cons of P/E, P/B, P/S, P/CF, D/P ratios

4.

Price-to-cashflow (P/CF) ratio

Price-to-cashflow ratio

$$P/CF = \frac{\text{market value of equity}}{\text{total cashflow (*)}} = \frac{\text{market price per share}}{\text{cashflow per share}}$$

(*) Cashflow used in CF calculation = net income + non-cash charges
Cash flow per share = EPS + non-cash charges per share.

Advantages and disadvantages

Advantages

- Cash flow is harder for managers to manipulate than earnings.
- Price to cash flow is more stable than price to earnings.
- Reliance on cash flow rather than earnings handles the problem of differences in the quality of reported earnings, which is a problem for P/E.
- Differences in price to cash flow are significantly related to differences in long-run average stock returns.

Disadvantages

- Items affecting actual cash flow from operations are ignored when the EPS plus noncash charges estimate is used. For example, noncash revenue and net changes in working capital are ignored.
- From a theoretical perspective, free cash flow to equity (FCFE) is preferable. However, FCFE is more volatile than operating cash flow, so it is not necessarily more informative.

[LOS 4.b.c.d] Calculate and determine the pros and cons of P/E, P/B, P/S, P/CF, D/P ratios

4.

Dividend yield

Trailing D/P

$$\text{trailing D/P} = \frac{4 \times \text{most recent quarterly dividend}}{\text{Market price per share}}$$

Leading D/P

$$\text{leading D/P} = \frac{\text{forecasted dividends over next four quarters}}{\text{market price per share}}$$

Advantages and disadvantages

Advantages

- Dividend yield contributes to total investment return.
- Dividends are not as risky as the capital appreciation component of total return.

Disadvantages

- The focus on dividend yield is incomplete because it ignores capital appreciation.
- The dividend displacement of earnings concept argues that dividends paid now displace future earnings, which implies a trade-off between current and future cash flows.

[LOS 4.b.c.d] Calculate and determine the pros and cons of P/E, P/B, P/S, P/CF, D/P ratios

3.

Dividend yield

Example: Calculating dividend yield

OnePrice, Inc., just paid a dividend of \$0.50 per share. The consensus forecasted dividends for OnePrice, Inc., over the next four quarters are \$0.50, \$0.55, \$0.60, and \$0.65. The current market price is \$47.50. Calculate the leading and trailing dividend yield.

Answer:

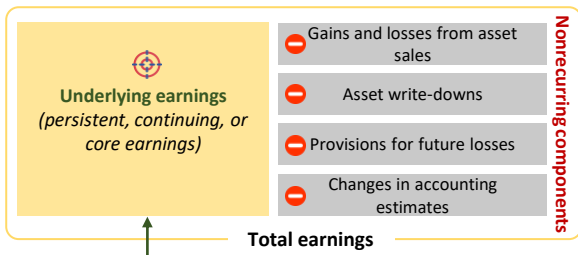
$$\text{Trailing D/P} = \frac{4 \times \text{most recent quarterly dividend}}{\text{Market price per share}} = \frac{4 \times \$0.5}{\$47.5} = 0.042 = 4.2\%$$

$$\begin{aligned} \text{Leading D/P} &= \frac{\text{forecasted dividends over next four quarters}}{\text{market price per share}} \\ &= \frac{\$0.5 + \$0.55 + \$0.60 + \$0.65}{\$47.5} = \frac{\$2.3}{\$47.5} = 0.048 = 4.8\% \end{aligned}$$

[LOS 4.e] Calculate and interpret underlying earnings, explain methods of normalizing earnings per share (EPS), and calculate normalized EPS.

Estimating the appropriate earnings measure is crucial to successfully using the P/E ratio in market-based valuation.

1. Underlying earnings



Key focus of an analyst is estimating **underlying earnings**

Molodovsky effect

Earnings contain a transitory portion that is due to cyclical.

Molodovsky effect: The countercyclical tendency to have high P/E's due to lower EPS at the bottom of the cycle and low P/E's due to high EPS at the top of the cycle.

[LOS 4.e] Calculate and interpret underlying earnings, explain methods of normalizing earnings per share (EPS), and calculate normalized EPS.

2.

Normalized earnings

Analysts adjust P/Es for cyclicalities by estimating **normalized (or normal) earnings per share**, which is an estimate of EPS in the middle of the business cycle.

The following two methods are used to normalize earnings:

Historical average EPS method

Straightly calculate the average EPS over the most recent business cycle.

Average return on equity method

Step 1: Calculate the average ROE over the most recent business cycle.

Step 2: Calculate average EPS by multiplying average return on equity (ROE) by the current book value per share (BVPS).

Normalised EPS = average ROE x current BVPS

[LOS 4.e] Calculate and interpret underlying earnings, explain methods of normalizing earnings per share (EPS), and calculate normalized EPS.

2.

Normalized earnings

Example: Calculating normalized earnings

Using the data in the following figure, calculate normalized earnings using the method of historical average EPS and the method of average return on equity for Magnolia Enterprises.

Data for Magnolia Enterprises [amounts in Canadian dollars (C\$)]

Year	2020	2021	2022	2023
EPS	C\$4.20	C\$3.75	C\$4.75	C\$4.30
BVPS	C\$26.02	C\$27.78	C\$29.25	C\$32.29
ROE	14%	12.0%	16.0%	14%

[LOS 4.e] Calculate and interpret underlying earnings, explain methods of normalizing earnings per share (EPS), and calculate normalized EPS.

2.
Normalized earnings

Example: Calculating normalized earnings

Answer:

1. Historical average EPS method:

Normalised earnings = average EPS over the most recent business cycle

$$= \frac{4.2 + 3.75 + 4.75 + 4.30}{4} = \text{C\$}4.25$$

2. Average return on equity method:

$$\text{Average ROE} = \frac{0.14 + 0.12 + 0.16 + 0.14}{4} = 0.14 = 14\%$$

Normalised EPS = average ROE x current BVPS = 0.14 x C\$32.29 = C\$4.52

Normalized earnings are C\$4.25 based on the method of historical average EPS and C\$4.52 based on the method of average return on equity.

[LOS 4.f] Explain and justify the use of earnings yield (E/P).

Analysts adjust EPS for differences in accounting methods between the company and companies it is being compared with so that the P/Es will be comparable.

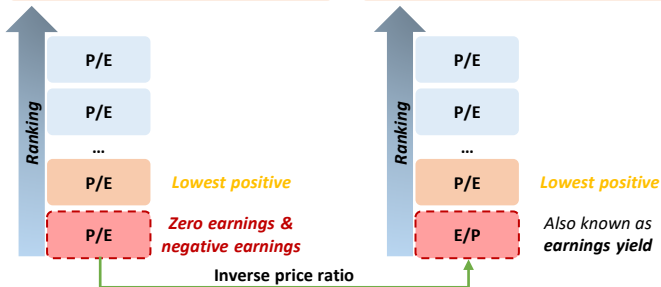
Using price multiples method



Problem



Solution



In case of negative earnings render P/E ratios meaningless, we use normalized EPS and/or restate the ratio as the earnings yield (E/P) because price is never negative.

Implication: A high E/P suggests a cheap security, and a low E/P suggests an expensive security, so securities can be ranked from cheap to expensive based on E/P ratios.

[LOS 4.g] Describe fundamental factors that influence alternative price multiples and dividend yield.

Price multiples may also be applied to valuation in the method based on forecasted fundamentals.

**1. Justified P/E
multiple**

**2. Justified P/B
multiple**

**3. Justified P/S
multiple**

**4. Justified P/CF
multiple**

**5. Justified
EV/EBITDA multiple**

**6. Justified
dividend yield**

[LOS 4.i] Calculate and interpret the justified price-to-earnings ratio (P/E), price-to-book ratio (P/B), and price-to-sales ratio (P/S) for a stock, based on forecasted fundamentals.

1. Justified P/E multiple

To understand fundamental price multiples, consider the Gordon growth valuation model:

$$P_0 = \frac{D_1}{r - g} = \frac{D_0 \times (1 + g)}{r - g}$$

Case 1: Divide both sides of the equation by next year's earnings forecast, E_1 :

$$\frac{P_0}{E_1} = \frac{D_0 \times (1 + g)}{E_1 \times (r - g)}$$

Case 2: Divide both sides of the equation by previous year's earnings forecast, E_0 :

$$\frac{P_0}{E_0} = \frac{D_0 \times (1 + g)}{E_0 \times (r - g)}$$

If we express D as the product of current earnings per share (E) and the payout ratio (D/E) and express the retention rate as b

$$\frac{P_0}{E_1} = \frac{(1 - b)}{(r - g)}$$

This is referred to as a **leading P/E** ratio.

$$\frac{P_0}{E_0} = \frac{(1 - b) \times (1 + g)}{(r - g)}$$

This is referred to as a **lagging or trailing P/E** ratio.

This P/E based on fundamentals is referred to as a **justified P/E**.

[LOS 4.i] Calculate and interpret the justified price-to-earnings ratio (P/E), price-to-book ratio (P/B), and price-to-sales ratio (P/S)...

1.

Justified P/E multiple

Conclusion:

The fundamental factors that affect P/E are expected growth rate and required return (which is related to risk):

- Positively related to the growth rate of expected cash flows, whether defined as dividends or free cash flows, all else equal.
- Inversely related to the stock's required rate of return, all else equal.
- While not directly apparent, P/E ratio is positively related to inflation pass-through rate. If two similar companies operate in the same inflationary environment, the firm with the higher inflation pass-through rate (i.e., ability to pass some portion of higher costs on to customers) should have the higher P/E ratio.

[LOS 4.i] Calculate and interpret the justified price-to-earnings ratio (P/E), price-to-book ratio (P/B), and price-to-sales ratio (P/S)...

1.

Justified P/E multiple

 *Calculating justified P/E ratio.*

A stock has a payout ratio of 40%. The shareholders require a return of 11% on their investment, and the expected growth rate in dividends is 5%. Calculate the trailing and leading P/E multiple based on these forecasted fundamentals.

Answer:

$$\text{Justified leading P/E} = \frac{(1 - b)}{(r - g)} = \frac{0.4}{0.11 - 0.05} = 6.67$$

$$\text{Justified trailing P/E} = \frac{(1 - b) \times (1 + g)}{(r - g)} = \frac{0.4 \times 1.05}{0.11 - 0.05} = 7.00 \text{ or } 6.67 \times 1.05 = 7.00$$

[LOS 4.i] Calculate and interpret the justified price-to-earnings ratio (P/E), price-to-book ratio (P/B), and price-to-sales ratio (P/S)...

2. Justified P/B multiple

Consider the Gordon growth valuation model:

$$P_0 = \frac{D_1}{r - g}$$



①

Since $D_1 = E_1 \times (1 - b)$:

$$P_0 = \frac{E_1 \times (1 - b)}{r - g}$$

②

As $E_1 = B_0 \times \text{ROE}$:

$$P_0 = \frac{B_0 \times \text{ROE} \times (1 - b)}{r - g}$$

③

Divide both sides of the equation by B_0 :

$$\frac{P_0}{B_0} = \frac{B_0 \times \text{ROE} \times (1 - b)}{B_0 \times (r - g)}$$

④

Simplified the calculation, combining with $g = \text{ROE} \times b$:

$$\frac{P_0}{B_0} = \frac{\text{ROE} \times (1 - g/\text{ROE})}{r - g}$$



$$\frac{P_0}{B_0} = \frac{\text{ROE} - g}{r - g}$$

[LOS 4.i] Calculate and interpret the justified price-to-earnings ratio (P/E), price-to-book ratio (P/B), and price-to-sales ratio (P/S)...

2.

Justified P/B multiple

Conclusion:

The fundamental factors that affect P/B:

- P/B increases as ROE increases, all else equal.
- The larger the spread between ROE and r , all else equal, the higher the P/B ratio. This makes sense if you remember that ROE is the return on the firm's investment projects and r is the required return. The larger the spread, all else equal, the more value the firm is creating through its investment activities and the higher its market value as represented by V_0

[LOS 4.i] Calculate and interpret the justified price-to-earnings ratio (P/E), price-to-book ratio (P/B), and price-to-sales ratio (P/S)...

3. Justified P/S multiple

Consider the Gordon growth valuation model:

$$P_0 = \frac{D_1}{r - g} = \frac{D_0 \times (1 + g)}{r - g}$$



Since $D_0 = E_0 \times (1 - b)$:

$$P_0 = \frac{E_0 \times (1 - b) \times (1 + g)}{r - g}$$

Divide both sides of the equation by S_0 :

$$\frac{P_0}{S_0} = \frac{E_0 \times (1 - b) \times (1 + g)}{S_0 \times (r - g)}$$



Since E_0 / S_0 is known as Net profit margin:

$$\frac{P_0}{S_0} = \text{Net profit margin} \times \text{Justified trailing P/E}$$

Besides, net profit margin influences indirectly through its effect on the sustainable growth rate, g :

$$g = b \times \text{Net profit margin} \times \frac{\text{Sales}}{\text{Assets}} \times \frac{\text{Assets}}{\text{Shareholders' Equity}}$$

[LOS 4.i] Calculate and interpret the justified price-to-earnings ratio (P/E), price-to-book ratio (P/B), and price-to-sales ratio (P/S)...

4.

Justified P/CF multiple

The justified price to cash flow based on fundamentals can be calculated by:

- Finding the value of the stock using a DCF model.
- Dividing the result by the chosen measure of cash flow.

P/CF **positively** related to growth rate

P/CF **negatively** related to required return

[LOS 4.i] Calculate and interpret the justified price-to-earnings ratio (P/E), price-to-book ratio (P/B), and price-to-sales ratio (P/S)...

5.

Justified EV/EBITDA multiple

EV = market value of common and preferred stock + market value of debt – cash and short-term investment

The justified EV/EBITDA based on fundamentals is simply the enterprise value based on a forecast of fundamentals divided by EBITDA forecast based on fundamentals.

EV/EBITDA **positively** related to growth rate in FCFF and EBITDA

P/CF **negatively** related to firm's overall risk level and weighted average cost of capital (WACC)

[LOS 4.i] Calculate and interpret the justified price-to-earnings ratio (P/E), price-to-book ratio (P/B), and price-to-sales ratio (P/S)...

6. Justified dividend yield

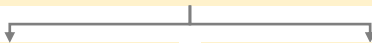
Consider the Gordon growth valuation model:

$$P_0 = \frac{D_1}{r - g} = \frac{D_0 \times (1 + g)}{r - g}$$



Cross multiplication, we have:

$$\frac{P_0}{D_0} = \frac{r - g}{1 + g}$$



Dividend yield **positively** related to required return

P/CF **negatively** related to forecasted growth rate in dividends



This implies that choosing high dividend yield stocks reflects a value rather than a growth investment strategy.

[LOS 4.h] Calculate and interpret a predicted P/E, given a cross-sectional regression on fundamentals, and explain limitations to the cross-sectional regression methodology

1.

Calculating predicted P/E

Cross-sectional regression refers to a regression in which the explained and explanatory variables are all based on the same point in time.

A **predicted P/E** can be estimated from linear regression of historical P/Es on its fundamental variables, including expected growth and risk.

While such empirical analysis can provide an analyst with useful insight, there are three main limitations:

- The predictive power of the estimated P/E regression for a different time period and/or sample of stocks is uncertain.
- The relationships between P/E and the fundamental variables examined may change over time.
- Multicollinearity is often a problem in these time series regressions, which makes it difficult to interpret individual regression coefficients.

[LOS 4.h] Calculate and interpret a predicted P/E, given a cross-sectional regression on fundamentals, and...

1.

Calculating predicted P/E

Example: Calculating predicted P/E

An analyst is valuing a public utility with a dividend payout ratio of 0.50, a beta of 0.95, and an expected earnings growth rate of 0.06. A regression on other public utilities produces the following regression equation:

$$\text{Predicted P/E} = 6.75 + 4.00 \times \text{dividend payout} + 12.35 \times \text{growth} - 0.5 \times \text{beta}$$

The firm's P/E ratio is 12.0. Calculate the predicted P/E on the basis of the values of the explanatory variables for the company, and determine whether the stock is over or underpriced.

Answer:

We have:

$$\begin{aligned} \text{Predicted P/E} &= 6.75 + 4.00 \times \text{dividend payout} + 12.35 \times \text{growth} - 0.5 \times \text{beta} \\ &= 6.75 + 4.00 \times 0.50 + 12.35 \times 0.06 - 0.5 \times 0.95 \\ &= 9.02 \end{aligned}$$

Actual P/E is greater than predicted P/E, so the firm is overpriced.

[LOS 4.I,m] Evaluate whether a stock is overvalued, fairly valued, or undervalued based on comparisons of multiples; Evaluate a stock by the method of comparables and explain the importance of fundamentals in using the method of comparables

1.**Evaluating value of a stock**

The basic idea of the method of comparables is to compare a stock's price multiple to that of a benchmark portfolio:

- Firms with multiples below the benchmark are undervalued
- Firms with multiples above the benchmark are overvalued

Problems: The fundamentals of the stock may differ from the fundamentals of the benchmark → We can not make direct comparisons and draw any conclusions about whether the stock is overvalued or undervalued.

[LOS 4.I,m] Evaluate whether a stock is overvalued, fairly valued, or undervalued based on comparisons of multiples...

1.

Evaluating value of a stock

*Let's use the P/E ratio as an example. Remember that justified P/E is positively related to growth rates and negatively related to required rate of return and risk. Suppose we determine that the P/E of our stock is less than the benchmark. There are (at least) **three** possible explanations for this:*

01

The stock is undervalued.

02

The stock is properly valued, but the stock has a lower expected growth rate than the benchmark, which leads to a lower P/E.

03

The stock is properly valued, but it has a higher required rate of return (higher risk) than the benchmark, which leads to a lower P/E.

In order to conclude that the stock is truly undervalued, we have to make sure that the stock is comparable to the benchmark; it should have **similar expected growth** and **similar risk**.

[LOS 4.I,m] Evaluate whether a stock is overvalued, fairly valued, or undervalued based on comparisons of multiples...

1.

Evaluating value of a stock

Example: Evaluating P/E ratios with the method of comparables

An analyst has gathered P/E information on two stocks, Allbright Interiors and Basic Designs.

Market Data on Allbright Interiors and Basic Designs

	Trailing P/E	Leading P/E	5-year growth rate	Beta
Allbright	10.0	8.7	11.0%	1.3
Basic Designs	14.0	12.7	9.0%	1.4
Peer median	13.3	12.1	11.0%	1.3

Evaluate the value and P/E of each stock based on the method of comparables.

Answer:

Allbright has a lower P/E than the peer median, despite the fact that it has a comparable growth rate and beta. This indicates Allbright is undervalued. Basic Designs, on the other hand, has a higher P/E, despite lower expected growth and a higher beta, which suggests it's overvalued relative to the benchmark.

[LOS 4.I,m] Evaluate whether a stock is overvalued, fairly valued, or undervalued based on comparisons of multiples...

1.

Evaluating value of a stock

- The same steps used in valuing stocks with P/E's apply when using P/Bs.
 - The *major difference* between the approaches is that book value forecasts are not widely disseminated like they are for EPS.
 - Most analysts use *trailing book values* in calculating P/Bs.
 - Relative P/B valuation must consider differences in ROE, risk, and expected growth in making comparisons among stocks.
-
- P/S valuation using the method of comparables follows the same steps as for P/E and P/B.
 - P/S ratios are usually calculated based on *trailing sales*.
 - Analysts need to control for profit margin, expected growth, risk, and the quality of accounting data in making comparisons.

[LOS 4.I,m] Evaluate whether a stock is overvalued, fairly valued, or undervalued based on comparisons of multiples...

1.

Evaluating value of a stock

Example: Evaluating P/B and P/S ratios with the method of comparables

Cisco Systems belongs to the Networking Products industry group, and Soothsayer belongs to the Enterprise Software/Services industry group. Recall that the P/B ratios for Cisco and Soothsayer were 4.45 and 10.04, respectively, and the P/S ratios were 6.60 and 6.71. Determine whether the two stocks are overvalued or undervalued compared to their peer group means and medians.

Basic Data From the Computer Industry

Peer group	Mean P/B	Median P/B	Mean P/S	Median P/S
Networking products	2.065	1.170	3.733	0.900
Enterprise software/services	7.866	2.770	3.341	1.920

[LOS 4.I,m] Evaluate whether a stock is overvalued, fairly valued, or undervalued based on comparisons of multiples...

1.

Evaluating value of a stock

Example: Evaluating P/B and P/S ratios with the method of comparables

Answer:

The P/B ratio for Crisco Systems exceeds the mean P/B ratio for the peer group (2.065) as well as the median P/B ratio (1.170) for the peer group; therefore, by this measure the stock would appear to be overvalued. The P/S ratio also exceeds both the mean P/S (3.733) and the median P/S (0.900) for the peer group, which also indicates that the stock is overvalued.

The P/B ratio for Soothsayer exceeds the peer group mean P/B (7.866) as well as the peer group median P/B (2.770) and suggests that the stock is overvalued. Similarly, the P/S ratio for Soothsayer exceeds the peer group mean P/S (3.341) as well as the peer group median P/S (1.920) and indicates that Soothsayer stock is overvalued as well.

Note the significant disparity between the mean and median values for each peer group. This is a clear indication of the presence of outliers in the data.

[LOS 4.I,m] Evaluate whether a stock is overvalued, fairly valued, or undervalued based on comparisons of multiples...

1.

Evaluating value of a stock

In line with other valuations by comparables discussed earlier (P/E, P/B, and P/S), a *lower* EV/EBITDA relative to peer firms indicates relative *undervaluation*, and vice versa, everything else being equal.

The process for dividend yield is similar to that for other multiples. This assumes that the peers have been identified on the basis of comparable risk and growth.

→ *High* dividend yield relative to the benchmark indicates *undervaluation*, all else equal.

[LOS 4.I,m] Evaluate whether a stock is overvalued, fairly valued, or undervalued based on comparisons of multiples...

2.

The Fed and Yardeni Models

The **Fed model** considers the overall market to be overvalued (undervalued) when the earnings yield (i.e., the E/P ratio) on the S&P 500 Index is lower (higher) than the yield on 10-year U.S. Treasury bonds.

The **Yardeni model** includes expected earnings growth rate in the analysis:

$$CEY = CBY - k \times LTEG + \epsilon_t$$

where:

CEY = current earnings yield of the market

CBY = current Moody's A-rated corporate bond yield

LTEG = five-year consensus earnings growth rate

k = constant assigned by the market to earnings growth (about 0.20 in recent years)

Taking reciprocals of the Yardeni model (and ignoring the error term), we get the P/E ratio that is negatively related to interest rates and positively related to growth.

$$\frac{P}{E} = \frac{1}{CBY - k \times LTEG}$$

[LOS 4.j] Calculate and interpret the P/E-to-growth (PEG) ratio and explain its use in relative valuation

1.

The P/E-to-growth (PEG) ratio

The relationship between earnings growth and P/E is captured by the **P/E-to-growth (PEG) ratio**:

PEG ratio

$$\text{PEG ratio} = \frac{\text{P/E ratio}}{g}$$

The PEG is interpreted as P/E per unit of expected growth.

Remember that the growth rate is one of the fundamental factors that affect P/E (P/E is directly related to the growth rate).

The PEG ratio, in effect, “standardizes” the P/E ratio for stocks with different expected growth rates.

The implied valuation rule: stocks with lower PEGs are more attractive than stocks with higher PEGs, assuming that risk is similar.

[LOS 4.j] Calculate and interpret the P/E-to-growth (PEG) ratio and explain its use in relative valuation

1.

The P/E-to-growth (PEG) ratio

There are a number of drawbacks to using the PEG ratio:

01

The relationship between P/E and g is not linear, which makes comparisons difficult.

02

The PEG ratio still doesn't account for risk.

03

The PEG ratio doesn't reflect the duration of the high-growth period for a multistage valuation model, especially if the analyst uses a short-term high-growth forecast.

[LOS 4.j] Calculate and interpret the P/E-to-growth (PEG) ratio and explain its use in relative valuation

1.

The P/E-to-growth (PEG) ratio

Example: Calculating and using the PEG ratio

Med-Ready, Inc., has a leading P/E of 28.75 and a 5-year consensus growth rate forecast of 14.5%. The median PEG for a group of companies comparable in risk to Med-Ready is 2.34. Calculate the firm's PEG and explain whether the stock appears to be correctly valued, overvalued, or undervalued.

Answer:

The firm's PEG is:

$$\begin{aligned}\text{PEG ratio} &= \frac{\text{P/E ratio}}{g} \\ &= \frac{28.75}{14.5} \\ &= 1.98.\end{aligned}$$

Given the comparable group median PEG of 2.34, it appears that Med-Ready may be undervalued.

However, it is important for the analyst to determine whether the peer group PEG is also based on leading P/Es and whether the comparable firms are similar in risk.

[LOS 4.k] Calculate and explain the use of price multiples in determining terminal value in a multistage discounted cash flow (DCF) model

1.

Calculating terminal value with price multiples

A **terminal value** that is projected as of the end of the investment horizon should reflect the earnings growth that a firm can sustain over the long run, beyond that point in time.

Analysts often use terminal price multiples like P/E, P/B, P/S, and P/CF to estimate terminal value.

There are **two** methods of estimating the price multiple: based on **fundamentals** and based on **comparables**.

Based on fundamentals

Terminal value in year n

(Justified leading P/E ratio) $\times$
(forecasted earnings in year $n + 1$)
or
(Justified trailing P/E ratio) $\times$
(forecasted earnings in year n)

Based on comparables

Terminal value in year n

(Benchmark leading P/E ratio) $\times$
(forecasted earnings in year $n + 1$)
or
(Benchmark trailing P/E ratio) $\times$
(forecasted earnings in year n)

[LOS 4.k] Calculate and explain the use of price multiples in determining terminal value in a multistage...

1.

Calculating terminal value with price multiples

There are **two** methods of estimating the price multiple: based on **fundamentals** and based on **comparables**.

Based on fundamentals

Strength: It is not as dependent on the selection of peer companies, as it focuses on the target company's financial data.

Weakness: It requires estimates of the growth rate, required rate of return, and payout ratio.

Based on comparables

Strength: It uses market data exclusively and is more appropriate for assessing a company's relative valuation within its industry.

Weakness: A benchmark marred by mispricing will transfer that error to the estimated terminal value.

[LOS 4.k] Calculate and explain the use of price multiples in determining terminal value in a multistage...

1.

Calculating terminal value with price multiples

Example: Calculating terminal value with price multiples

An analyst estimates the EPS of Polar Technology in five years to be C\$2.10, the EPS in six years to be C\$2.32, and the median trailing industry P/E to be 35. Calculate the terminal value in Year 5.

Answer:

We have:

Terminal value in Year 5 = (Benchmark trailing P/E ratio) x (forecasted earnings in year 5)

$$= 35 \times \text{C\$}2.10$$

$$= \text{C\$}73.50.$$

[LOS 4.n] Explain alternative definitions of cash flow used in price and enterprise value (EV) multiples and describe limitations of each definition

1.

Alternative definitions of cash flow

*There are at least **four** definitions of cash flow available for use in calculating the P/CF ratio:*

- (1) earnings-plus-noncash-charges (CF);*
- (2) adjusted cash flow (adjusted CFO);*
- (3) free cash flow to equity (FCFE); and*
- (4) earnings before interest, taxes, depreciation, and amortization (EBITDA).*

1.1. Earnings-plus-noncash-charges (CF):

$$CF = \text{Net income} + \text{Depreciation} + \text{Amortization}$$

Limitation: It ignores some items that affect cash flow, such as noncash revenue and changes in net working capital.

[LOS 4.n] Explain alternative definitions of cash flow used in price and enterprise value (EV) multiples and...

1.

Alternative definitions of cash flow

1.2. Adjusted cash flow (adjusted CFO):

CFO is often adjusted for nonrecurring cash flows.

US GAAP

Requires interest paid, interest received, and dividends received to be classified as operating cash flows.

IFRS

Interest paid may be classified as either an operating or financing cash flow, while interest and dividends received can be classified as either operating or investing cash flows.

Care should be taken when comparing firms reporting under different standards.

[LOS 4.n] Explain alternative definitions of cash flow used in price and enterprise value (EV) multiples and...

1.

Alternative definitions of cash flow

1.3. Free cash flow to equity (FCFE):

Theory suggests that FCFE is the preferred way to define cash flow, but it is more volatile than straight cash flow.

$$\text{FCFE} = \text{CFO} - \text{FCInv} + \text{Net borrowing}$$

where:

FCInv is Fixed capital investment

Net borrowing = (Long- and short-term debt issues) – (Long- and short-term repayments).

[LOS 4.n] Explain alternative definitions of cash flow used in price and enterprise value (EV) multiples and...

1.

Alternative definitions of cash flow

1.4. Earnings before interest, taxes, depreciation, and amortization (EBITDA):

EBITDA is a pretax, pre-interest measure that represents a flow to both equity and debt.

→ It is better suited as an indicator of total company value than just equity value.

More on this point is provided in our discussion of the enterprise value (EV)-to-EBITDA ratio.

Analysts typically use *trailing* price to cash flow ratio, which relies on the most recent four quarters of cash flow per share.

Given one of the four definitions of cash flow, the P/CF ratio is calculated as:

P/CF ratio

$$\text{P/CF ratio} = \frac{\text{Market value of equity}}{\text{cash flow}} = \frac{\text{Market price per share}}{\text{cash flow per share}}$$

where:

cash flow = CF, adjusted CFO, FCFE, or EBITDA

[LOS 4.n] Explain alternative definitions of cash flow used in price and enterprise value (EV) multiples and...

1.

Alternative definitions of cash flow

Example: Calculating P/CF

Data Management Systems, Inc., (DMS) reported net income of \$32 million, depreciation and amortization of \$41 million, net interest expense of \$12 million, and cash flow from operations of \$44 million. The tax rate is 30%. Calculate the P/CF ratio using earnings-plus-non-cash-charges (CF) as a proxy for cash flow. DMS has 25 million shares of common stock outstanding, trading at \$47 per share.

Answer:

Using earnings-plus-non-cash-charges (CF) as a proxy for cash flow, we have:

CF = Net income + Depreciation + Amortization

= \$32 million + \$41 million

= \$73 million .

Market value of equity = 25 million shares x \$47 per share

= \$1,175 million.

Thus, $P/CF = \frac{\text{Market value of equity}}{\text{cash flow}}$

= $\frac{1,175}{73}$

= 16.1 times.

[LOS 4.o] Calculate and interpret EV multiples and evaluate the use of EV/EBITDA

1.

Calculating EV multiples

As discussed, because EBITDA is a flow to both equity and debt, it should be related to a numerator that measures total company value.

Enterprise value (EV) is total company value:

EV calculation

EV = Market value of common stock + market value of preferred equity + market value of debt + minority interest – cash and investments

The rationale for subtracting cash and investments is that an acquirer's net price paid for an acquisition target would be lowered by the amount of the target's liquid assets.

EV/EBITDA is the ratio of enterprise value to EBITDA:

EV/EBITDA calculation

$$\text{EV/EBITDA} = \frac{\text{Enterprise value}}{\text{EBITDA}}$$

[LOS 4.o] Calculate and interpret EV multiples and evaluate the use of EV/EBITDA

2.

Evaluating the use of EV/EBITDA

Advantage

The ratio may be more useful than P/E when comparing firms with different degrees of financial leverage.

EBITDA is useful for valuing capital-intensive businesses with high levels of depreciation and amortization.

EBITDA is usually positive even when EPS is not.

Drawback

If working capital is growing, EBITDA will overstate CFO. Further, the measure ignores how different revenue recognition policies affect CFO.

Because FCFF captures the amount of capital expenditures, it is more strongly linked with valuation theory than EBITDA. EBITDA will be an adequate measure if capital expenses equal depreciation expenses.

[LOS 4.o] Calculate and interpret EV multiples and evaluate the use of EV/EBITDA

3.

An alternative measure of a company's overall value

An alternative measure of a company's overall value is **total invested capital (TIC)**, sometimes referred to as **market value of invested capital**.

- Total invested capital is the market value of the company's equity and debt.
- Unlike enterprise value, TIC includes cash and short-term investments.

In addition to EV/EBITDA and TIC/EBITDA, analysts employ enterprise value ratios with EBIT, FCFF, or other items in the denominator.

Example: The enterprise value to sales (EV/S) ratio can be used as an alternative to the P/S ratio. The EV/S ratio is appropriate for comparing companies with significantly different capital structures.

[LOS 4.o] Calculate and interpret EV multiples and evaluate the use of EV/EBITDA

3.

An alternative measure of a company's overall value

Example: Calculating EV/EBITDA

An analyst gathered the following data for Boulevard Industries [all amounts in Swiss francs (Sf)]:

Recent share price	Sf 22.50
Share outstanding	40 million
Market value of debt	Sf 137 million
Cash and marketable securities	Sf 62.3 million
Investments	Sf 327 million
Net income	Sf 137.5 million
Interest expense	Sf 6.9 million
Depreciation and amortization	Sf 10.4 million
Taxes	Sf 95.9 million

Based on this information, calculate the EV/EBITDA ratio for Boulevard Industries.

[LOS 4.o] Calculate and interpret EV multiples and evaluate the use of EV/EBITDA

3.
An alternative measure of a company's overall value
Example: Calculating EV/EBITDA
Answer:

Recall: EBITDA = Net income + Interest expense + Taxes + Depreciation and amortization

$$= 137.5 + 6.9 + 95.9 + 10.4$$

$$= \text{Sf } 250.7 \text{ million}$$

EV = Market value of equity + Market value of debt - Cash and marketable securities - Investments

$$= 22.50 \times 40 + 137 - 62.3 - 327$$

$$= \text{Sf } 647.7 \text{ million}$$

$$\text{Thus, EV/EBITDA} = \frac{647.7}{250.7} = 2.6 \text{ times.}$$

[LOS 4.p] Explain sources of differences in cross-border valuation comparisons

1.

Sources of differences

Using relative valuation methods that require the use of comparable firms is challenging in an international context because:

- Comparing companies across borders frequently involves differences in ***accounting methods, cultures, risk*** and ***growth opportunities***.
- P/Es for individual companies in the same industry but in different countries have been found to vary widely.
- P/Es of different national markets often vary substantially at any single point in time.

Common differences in international accounting treatment fall into several categories:

- Goodwill
- Deferred income taxes
- Foreign exchange adjustments
- R&D
- Pension expense
- Tangible asset revaluations

Note: The *least* affected multiples due to *differences in accounting methods* is **P/FCFE**.

[LOS 4.q] Describe momentum indicators and their use in valuation

1. Momentum indicators

Momentum indicators relate either the market price or a fundamental variable (such as EPS) to the time series of historical or expected value.

Common momentum indicators include:

- (1) *Earnings surprise*
- (2) *Standardized unexpected earnings*
- (3) *Relative strength*

1.1. Earnings surprise:

Unexpected earnings or **earnings surprise** is the difference between reported earnings and expected earnings.

$$\text{Earnings surprise} = \text{Reported EPS} - \text{Expected EPS}$$

→ *This is usually scaled by a measure that expresses the variability of analysts' EPS forecasts.*

The economic rationale: positive surprises may lead to persistent positive abnormal returns.

[LOS 4.q] Describe momentum indicators and their use in valuation

1. Momentum indicators

1.2. Standardized unexpected earnings:

Similarly, the standardized unexpected earnings (SUE) measure is defined as:

$$\text{SUE} = \frac{\text{Earnings surprise}}{\text{Standard deviation of earnings surprise}}$$

A given size forecast error is more meaningful the smaller the size of the historical forecast errors.

1.3. Relative strength:

Relative strength indicators compare a stock's price or return performance during a given time period with its own historical performance or with some group of peer stocks.

The economic rationale: patterns of persistence or reversal may exist in stock returns.

[LOS 4.r] Explain the use of the arithmetic mean, the harmonic mean, the weighted harmonic mean, and the median to describe the central tendency of a group of multiples

1.

Use of means

We might be familiar with the arithmetic mean, now we will become acquainted with the weighted harmonic mean.

$$\text{Weighted harmonic mean} = \frac{1}{\sum_{i=1}^n \frac{w_i}{X_i}}$$

where w_i is the weight of X_i

Example: Consider two stocks: one priced at \$10 with earnings of \$1 per share ($P/E = 10$) and one priced at \$16 with earnings of \$2 ($P/E = 8$).

For a portfolio with one share of each stock, earnings per share are $1 + 2 = 3$ and the “price” of a portfolio share is $10 + 16 = 26$. The portfolio price-to-earnings is $26 / 3 = 8.67$.

We will demonstrate that the portfolio or index P/E (as well as other relative value ratios based on price) is not equal to the mean or weighted mean of the P/E s of the portfolio stocks, but is best calculated as the weighted harmonic mean P/E .

[LOS 4.r] Explain the use of the arithmetic mean, the harmonic mean, the weighted harmonic mean, and...

1.

Use of means

Consider the following alternative measures of the mean P/E for the portfolio:

- Arithmetic mean = $(8 + 10)/2 = 9$
- Weighted mean = $(10/26) \times 10 + (16/26) \times 8 = 8.76$
- Harmonic mean = $\frac{2}{\frac{1}{10} + \frac{1}{8}} = 8.88$
- Weighted harmonic mean = $\frac{1}{\frac{10}{26} \times \frac{1}{10} + \frac{16}{26} \times \frac{1}{8}} = 8.67$

- When there are extreme (high or low) outliers, the arithmetic mean will be the most affected.
- The harmonic mean puts more weight on smaller values. In this case, the median or weighted harmonic mean with the outliers excluded may be the most appropriate measures of the P/E for a portfolio or index.
- For an equal weighted portfolio or index, the harmonic mean and weighted harmonic mean will be equal.

MODULE 5 : Residual Income Valuation

LEARNING OUTCOMES

- 5.a.** Calculate and interpret residual income, economic value added, and market value added
- 5.b.** Describe the uses of residual income models
- 5.c.** Calculate the intrinsic value of a common stock using the residual income model and compare value recognition in residual income and other present value models
- 5.d.** Explain fundamental determinants of residual income
- 5.e.** Explain the relation between residual income valuation and the justified price-to-book ratio based on forecasted fundamentals
- 5.f.** Calculate and interpret the intrinsic value of a common stock using single-stage (constant-growth) and multistage residual income models
- 5.g.** Calculate the implied growth rate in residual income, given the market price-to-book ratio and an estimate of the required rate of return on equity
- 5.h.** Explain continuing residual income and justify an estimate of continuing residual income at the forecast horizon, given company and industry prospects

LEARNING OUTCOMES

- 5.i.** Compare residual income models to dividend discount and free cash flow models
- 5.j.** Explain strengths and weaknesses of residual income models and justify the selection of a residual income model to value a company's common stock
- 5.k.** Describe accounting issues in applying residual income models

[LOS 5.a] Calculate and interpret residual income, economic value added, and market value added

1.

Residual Income

Residual income (RI), or economic profit, is the net income of a firm less a charge that measures stockholders' opportunity cost of capital.

→ The rationale for the residual income approach is that it recognizes the cost of equity capital in the measurement of income.

Sales



EBIT



Less: tax and interest expense
(*cost of debt*)

Net Income



Less: dividend or other equity
funding costs (*cost of equity*)

Residual Income

Accounting net income includes a cost of debt (i.e., interest expense), but does not reflect dividends or other equity capital-related funding costs.

→ Accounting income may overstate returns from the perspective of equity investors.

Residual income explicitly *deducts all capital costs*.

[LOS 5.a] Calculate and interpret residual income, economic value added, and market value added

1.

Residual Income

Example: Calculating residual income

Madeira Fruit Suppliers, Inc., (MFS) distributes fruit to grocery stores in large U.S. cities. The book value of its assets is \$1.4 billion, which is financed with \$800 million in equity and \$600 million in debt. Its before-tax cost of debt is 3.33%, and its marginal tax rate is 34%. MFS has a cost of equity of 12.3%. MFS's abbreviated income statement is shown in the following figure.

Partial Income Statement for MFS

EBIT	\$142,000,000
Less: Interest expense	(20,000,000)
Pretax income	122,000,000
Less: Income tax expense	(41,480,000)
Net income	\$80,520,000

Determine whether MFS is profitable by calculating residual income and explaining its relationship to reported accounting income.

[LOS 5.a] Calculate and interpret residual income, economic value added, and market value added

1.

Residual Income

Answer:

While the accounting net income of \$80,520,000 indicates that MFS is profitable, it remains to be seen whether the firm is profitable after deducting a charge for equity. The dollar-based equity charge is:

Equity charge = equity capital x cost of equity

$$= \$800,000,000 \times 0.123$$

$$= \$98,400,000$$

Residual income is calculated as:

$$RI = NI - \text{Equity charge}$$

$$= \$80,520,000 - \$98,400,000$$

$$= -\$17,880,000.$$

→ Even though MFS is profitable in the traditional accounting sense, it is economically unprofitable after taking into account the necessary charge to meet stockholders' opportunity cost of supplying capital to the company.

[LOS 5.a] Calculate and interpret residual income, economic value added, and market value added

2.

Economic value added

Economic value added (EVA) measures the value added for shareholders by management during a given year.

Economic value added (EVA)

$$\begin{aligned} \text{EVA} &= \text{NOPAT} - (\text{WACC} \times \text{total capital}) \\ &= \text{EBIT} \times (1 - t) - \$\text{WACC} \end{aligned}$$

where:

NOPAT = $\text{EBIT} \times (1 - t)$ is net operating profit after tax

WACC is after tax weighted average cost of capital in decimal terms

t is marginal tax rate

$\$WACC$ is dollar cost of capital

Total capital = net working capital + net fixed assets = book value of long-term debt + book value of equity

[LOS 5.a] Calculate and interpret residual income, economic value added, and market value added

2.

Economic value added

The analyst should make the following adjustments (if applicable) to the financial statements before calculating NOPAT and invested capital:

01

Capitalize and amortize research and development charges (rather than expense them), and add them back to earnings to calculate NOPAT.

02

Add back charges on strategic investments that will generate returns in the future.

03

Eliminate deferred taxes and consider only cash taxes as an expense.

04

Treat operating leases as capital leases and adjust nonrecurring items.

05

Add LIFO reserve to invested capital and add back change in LIFO reserve to NOPAT.

[LOS 5.a] Calculate and interpret residual income, economic value added, and market value added

3.

Market value added

Market value added (MVA) is the difference between the market value of a firm's long-term debt and equity and the book value of invested capital supplied by investors. It measures the value created by management's decisions since the firm's inception.

Market value added (MVA)

$$\text{MVA} = \text{Market value of the company} - \text{Accounting book value of total capital}$$

Note: these models, like EVA and MVA, usually apply the concept of residual income to the measurement of managerial effectiveness and executive compensation. However, for the exam we're most interested in the equity valuation applications of *residual income models*.

[LOS 5.a] Calculate and interpret residual income, economic value added, and market value added

3.

Market value added

Example: Calculating EVA and MVA

VBM, Inc., reports NOPAT of \$2,100, a WACC of 14.2%, and invested capital of \$18,000 at the beginning of the year and \$21,000 at the end of the year. The market price (year-end) of the firm's stock is \$25 per share, and VBM has 800 shares outstanding. The market value (year-end) of the firm's long-term debt is \$4,000. Calculate VBM's EVA and MVA.

Answer:

First, we calculate EVA:

- $\$WACC = WACC \times \text{total capital} = 0.142 \times \$18,000 = \$2,556$
- $EVA = NOPAT - \$WACC = \$2,100 - \$2,556 = -\456

Next, we calculate MVA:

- The market value of the company is the market value of the equity plus the market value of the debt:

$$\begin{aligned} \text{MV of company} &= \text{MV of equity} + \text{MV of long-term debt} = (\$25 \times 800) + \$4,000 \\ &= \$24,000 \end{aligned}$$

- The firm's MVA is: $MVA = \$24,000 - \$21,000 = \$3,000$

[LOS 5.b] Describe the uses of residual income models

1. The uses of residual income models

Residual income (RI) models have been used to value both individual stocks and stock indexes, and they have also been proposed as a method to measure goodwill impairment.

A company that is generating more income than its cost of obtaining capital.

Positive residual income

Creating value

A company that is not generating enough income to cover its cost of capital.

Negative residual income

Destroying value

All else equal, higher (lower) residual income should be associated with higher (lower) valuations.

[LOS 5.c] Calculate the intrinsic value of a common stock using the residual income model and compare value recognition in residual income and other present value models

1. The Residual income model

The residual income model of valuation analyzes the intrinsic value of equity as the sum of **two** components:

01 The current book value of equity

02 The present value of expected future residual income

Note: when the change is made from valuing total shareholders' equity to directly valuing an individual common share, **earnings per share** rather than net income is used.

According to the residual income model, the intrinsic value of common stock can be expressed as follows:

Residual income valuation

$$V_0 = B_0 + \sum_{t=1}^{\infty} \frac{RI_t}{(1+r)^t} = B_0 + \sum_{t=1}^{\infty} \frac{E_t - r \times B_{t-1}}{(1+r)^t} = B_0 + \sum_{t=1}^{\infty} \frac{(ROE - r) \times B_{t-1}}{(1+r)^t}$$

[LOS 5.c] Calculate the intrinsic value of a common stock using the residual income model and...

1.

The Residual income model

Residual income valuation

$$V_0 = B_0 + \sum_{t=1}^{\infty} \frac{RI_t}{(1+r)^t} = B_0 + \sum_{t=1}^{\infty} \frac{E_t - r \times B_{t-1}}{(1+r)^t} = B_0 + \sum_{t=1}^{\infty} \frac{(ROE - r) \times B_{t-1}}{(1+r)^t}$$

where:

B_0 is the current book value of equity

RI_t is the residual income per share in year t

r is the required return on equity

E_t is the expected EPS for year t

B_{t-1} is the book value per share in year $t - 1$

ROE is the expected return on new investments (expected return on equity)

[LOS 5.c] Calculate the intrinsic value of a common stock using the residual income model and...

1.

The Residual income model

Example 1: Forecasting residual income

Laura Kraft, CFA, was assigned the task of forecasting the residual income for Delilah Cosmetics, Inc., over the next two years. To accomplish this task, Kraft assembled the information provided in the following figure. Kraft used a required rate of return of 11%. Forecast Delilah's residual income for 2025 and 2026.

Delilah Data Forecast

Current market price	\$24.00
Current book value per share (December 31, 2024)	\$18.00
Consensus annual EPS estimates	
• December 31, 2025	\$2.05
• December 31, 2026	\$2.22
Dividend payout ratio 2025 and 2026	65%

[LOS 5.c] Calculate the intrinsic value of a common stock using the residual income model and...

1.

The Residual income model

Example 1: Forecasting residual income

Answer:

Delilah Residual Income Forecast

	FY 2025	FY 2026
Beginning book value (B_{t-1})	\$18.00	\$18.72
Earnings per share forecast (E_t)	2.05	2.22
Dividend forecast ($D_t = E_t \times$ payout ratio)	1.33	1.44
Forecast book value per share ($B_{t-1} + E_t - D_t$)	18.72	19.50
Equity charge per share ($r \times B_{t-1}$)	1.98	2.06
Per share RI_t ($E_t - r \times B_{t-1}$)	\$0.07	\$0.16

[LOS 5.c] Calculate the intrinsic value of a common stock using the residual income model and...

1.

The Residual income model

Example 2: Computing intrinsic value with a residual income model

Consolidated Pipe Products has a required rate of return of 14%. The current book value is C\$6.50. Earnings forecasts for 2025, 2026, and 2027 are C\$1.10, C\$1.00, and C\$0.95, respectively. Dividends in 2025 and 2026 are forecasted to be C\$0.50 and C\$0.60, respectively. The dividend in 2027 is a liquidating dividend, which means that Consolidated will pay out its entire book value in dividends and cease doing business at the end of 2027. Calculate the value of Consolidated's stock using the residual income model.

Answer:

The residual income forecast is shown in the following table, with calculated values in blue.

Consolidated Pipe Residual Income Forecast (*see next slide*)

[LOS 5.c] Calculate the intrinsic value of a common stock using the residual income model and...

1.

The Residual income model

Example 2: Computing intrinsic value with a residual income model

Answer:

	2025	2026	2027
Beginning book value (B_{t-1})	C\$6.50	C\$7.10	C\$7.50
Earnings per share forecast (E_t)	1.10	1.00	0.95
Dividend forecast ($D_t = E_t$ x payout ratio)	0.50	0.60	8.45
Forecast book value per share ($B_{t-1} + E_t - D_t$)	7.10	7.50	0.00
Equity charge per share ($r \times B_{t-1}$)	0.91	0.99	1.05
Per share RI_t ($E_t - r \times$ B_{t-1})	C\$0.19	C\$0.01	-C\$0.10

[LOS 5.c] Calculate the intrinsic value of a common stock using the residual income model and...

1.

The Residual income model

Example 2: Computing intrinsic value with a residual income model

Answer:

The intrinsic value of Consolidated Pipe Products is its current book value plus the present value of the future residual income forecasts:

$$V_0 = \text{C}\$6.50 + \frac{\text{C}\$0.19}{1.14} + \frac{\text{C}\$0.01}{1.14^2} - \frac{\text{C}\$0.10}{1.14^3} = \text{C}\$6.61$$

We can also use the cash flow function on our calculators to solve this problem and save ourselves a little time. Here are the keystrokes:



CF0 = 6.50

C01 = 0.19

C02 = 0.01

C03 = -0.10

I = 14

CPT → NPV = 6.61.

[LOS 5.c] Calculate the intrinsic value of a common stock using the residual income model and...

2.

Value recognition in residual income model

	Residual income model	Other present value models (DDM and FCFE model)
Value tends to be recognized...	Earlier	Later
Large portion of the estimated intrinsic value comes from...	The current book value	The present value of the expected terminal value
<i>But the uncertainty of the expected terminal value is usually greater than any of the other forecasted cash flows because it occurs several years in the future. Hence:</i>		
Model's forecast error	Lower	Higher

[LOS 5.d] Explain fundamental determinants of residual income

1. Fundamental determinants of residual income

 **Recall:** We learned about the general residual income model. It makes no assumptions regarding the long-term future earnings or dividend growth.

If we make the simplifying assumption of a constant dividend and earnings growth rate, we can develop a residual income model that highlights the fundamental drivers of residual income:

Residual income valuation

$$V_0 = B_0 + \frac{(ROE - r) \times B_0}{r - g}$$

This version of the residual income model is referred to as the **single-stage residual income valuation model**:

- The **first term** is the current book value, the value of the company's assets net of liabilities.
- The **second term** is the present value of the expected future residual income.

[LOS 5.d] Explain fundamental determinants of residual income

1. Fundamental determinants of residual income

We can use this relationship to identify the **fundamental drivers** of residual income:

- If $ROE = r \rightarrow V_0 = B_0$. When $ROE > r$, the firm will have positive residual income and will be valued at more than book value ($V_0 > B_0$).
- $\frac{(ROE - r) \times B_0}{r - g}$ is the additional value generated by the firm's ability to produce returns in excess of the cost of equity and, consequently, is the present value of a firm's expected economic profits (i.e., residual income).

Tobin's Q, which is the ratio of the market value of debt and equity to the replacement cost of total assets, is a related concept with the expected future residual income:

Tobin's Q calculation

$$Q = \frac{\text{market value of debt} + \text{market value of equity}}{\text{replacement cost of total assets}}$$

Note: Residual income is likely to approach zero over time, as competitive forces drive industry profit margins to normal levels. Thus, in practice, the single-stage model is modified to handle declining RI by forecasting continuing residual income.

[LOS 5.e] Explain the relation between residual income valuation and the justified price-to-book ratio based on forecasted fundamentals

1.

Relation between RI valuation and justified P/B

As with the DDM and FCFE models, residual income models can be used to estimate justified price multiples. Below is the formula of single-stage residual income valuation model:

Single-stage residual income valuation model

$$V_0 = B_0 + \frac{(ROE - r) \times B_0}{r - g}$$

Among the various market multiples, residual income models are most closely related to the price-to-book value (P/B) ratio:

If $ROE > r$

→ The second term $\left(\frac{(ROE - r) \times B_0}{r - g}\right) > 0$

→ $V_0 > B_0$

→ The justified P/B ratio > 1 .

[LOS 5.f] Calculate and interpret the intrinsic value of a common stock using single-stage (constant-growth) and multistage residual income models

1.

Calculating the intrinsic value using single-stage model

We have learnt about how to calculate the intrinsic value using single-stage model:

Single-stage residual income valuation model

$$V_0 = B_0 + \frac{(ROE - r) \times B_0}{r - g}$$

Now we will consider some examples to see the detailed application of the formula above.

Note: Multistage residual income models will be discussed in LOS h.

Example 1: Calculating value with a single-stage residual income model

Western Atlantic Railroad has a book value of \$23.00 per share. The company's return on new investments (ROE) is 14%, and its required return on equity is 12%. The dividend payout ratio is 60%. Calculate the value of the shares using a singlestage residual income model and the present value of expected economic profits.

[LOS 5.f] Calculate and interpret the intrinsic value of a common stock using single-stage (constant-growth) and...

1.

Calculating the intrinsic value using single-stage model

Example 1: Calculating value with a single-stage residual income model

Answer:

Applying the single-stage residual income model formula, we have:

$$V_0 = B_0 + \frac{(ROE - r) \times B_0}{r - g}$$

We are given $B_0 = \$23.00$, $ROE = 14\%$, $r = 12\%$. Hence, we need to calculate the growth rate (g) as follows:

$$g = \text{retention ratio} \times ROE$$

$$= (1 - 0.6) \times 0.14 = 0.056 = 5.6\%$$

Then, calculate intrinsic value using the single-stage model:

$$\begin{aligned} V_0 &= B_0 + \frac{(ROE - r) \times B_0}{r - g} \\ &= \$23.00 + \frac{(0.14 - 0.12) \times \$23.00}{0.12 - 0.056} \\ &= \$30.19. \end{aligned}$$

Clearly, the present value of the firm's expected economic profits is $\frac{(0.14 - 0.12) \times \$23.00}{0.12 - 0.056} = \$7.19$.

[LOS 5.f] Calculate and interpret the intrinsic value of a common stock using single-stage (constant-growth) and...

1.

Calculating the intrinsic value using single-stage model

Example 2: Western Atlantic Railroad valuation with Gordon growth model

Use the information in the previous example to calculate the value of Western Atlantic common stock using the Gordon growth model.

Answer:

Recall: We learned about the GGM formula:

$$V_0 = \frac{D_1}{r - g}$$

From the previous example, we calculated $g = 5.6\%$ and $r = 12\%$ is given. Hence, we need to calculate D_1 as follows:

$$\begin{aligned} D_1 &= E_1 \times \text{dividend payout ratio} = \text{ROE} \times B_0 \times \text{dividend payout ratio} \\ &= 14\% \times \$23.00 \times 60\% \\ &= \$1.932 \end{aligned}$$

Thus, using the GGM we have:

$$V_0 = \frac{D_1}{r - g} = \frac{\$1.932}{0.12 - 0.056} = \$30.19$$

Notice that this is the same estimate as in the previous example where we used the single-stage residual income model.

[LOS 5.g] Calculate the implied growth rate in residual income, given the market price-to-book ratio and an estimate of the required rate of return on equity

1.

Calculating the implied growth rate in residual income

We can rearrange the single-stage residual income valuation model and solve for the growth rate in terms of the other variables:

Single-stage residual income valuation model

$$V_0 = B_0 + \frac{(ROE - r) \times B_0}{r - g}$$

Implied growth rate in residual income

$$g = r - \frac{(ROE - r) \times B_0}{V_0 - B_0}$$

This expression can now be used to directly compute the market's expectations of residual income growth implied by the current market price under the assumption that intrinsic value is equal to market price.

[LOS 5.g] Calculate the implied growth rate in residual income, given the market price-to-book ratio and...

1.

Calculating the implied growth rate in residual income

Example: Calculating implied growth rate

You are considering the purchase of Tellis Telecommunications, Inc., which has a P/B ratio of 2.50. ROE is expected to be 13%, current book value per share is €8.00, and the cost of equity is 11%. Calculate the growth rate implied by the current P/B ratio.

Answer:

The P/B ratio of 2.50 and the current book value per share of €8.00 imply a current market price of €20.00 (8×2.50). This implies a growth rate of:

$$\begin{aligned} g &= r - \frac{(ROE - r) \times B_0}{V_0 - B_0} \\ &= 0.11 - \frac{(0.13 - 0.11) \times €8.00}{€20.00 - €8.00} \\ &= 0.0967 \end{aligned}$$

Thus, the growth rate implied by the current P/B ratio is 9.67%.

[LOS 5.h] Explain continuing residual income and justify an estimate of continuing residual income at the forecast horizon, given company and industry prospects

1.

Continuing residual income

With multistage models, we can simplify the RI model by using the same multistage approach we used for DDM and FCF models: *Making some simplifying assumptions about the pattern of residual income growth over the long term after years.*

Continuing residual income is the residual income that is expected over the long term.

The residual income will continue beyond a specified earnings horizon depending on the fortunes of the industry and the sustainability of a specific firm's competitive prospects over the longer term.

[LOS 5.h] Explain continuing residual income and justify an estimate of continuing residual income at...

1.

Continuing residual income

To simplify the model, we typically make one of the following assumptions about continuing residual income at the end of the short-term period:

01

Residual income is expected to persist at its current level forever.

02

Residual income is expected to drop immediately to zero.

03

Residual income is expected to decline over time as ROE falls to the cost of equity (in which case residual income is eventually zero).

04

Residual income is expected to decline to a long-run average level consistent with a mature industry.

The projected rate at which residual income is expected to fade over the life cycle of the firm is captured by a **persistence factor**, ω ($0 \leq \omega \leq 1$).

The strength of ω will depend partly on the sustainability of the firm's competitive advantage and the structure of the industry: The more sustainable the competitive advantage and the better the industry prospects, the higher the persistence factor.

[LOS 5.h] Explain continuing residual income and justify an estimate of continuing residual income at...

1.

Continuing residual income

 *What are the persistence factors associated with*

Higher persistence factors will be associated with the following:

- Low dividend payouts
- Historically high residual income persistence in the industry

Lower persistence factors will be associated with the following:

- High return on equity
- Significant levels of nonrecurring items
- High accounting accruals

[LOS 5.h] Explain continuing residual income and justify an estimate of continuing residual income at...

1.

Continuing residual income

In the residual income model, intrinsic value is the sum of **three** components:

Multistage residual income valuation model

$$V_0 = B_0 + \text{PV of interim high-growth RI} + \text{PV of continuing RI}$$

We have **three** steps involved in the calculation as follows:

01 Calculate the current book value per share.

02 Calculate residual income in each year 1 to $T - 1$ during the interim highgrowth period and discount them back to today at the required return on equity.

03 Calculate continuing residual income that begins at the end of the high growth period starting in year T , and then calculate the present value of continuing residual income as of the end of year $T - 1$.

[LOS 5.h] Explain continuing residual income and justify an estimate of continuing residual income at...

1.

Continuing residual income

PV of continuing RI in year $T - 1$

$$PV = \frac{RI_T}{1 + r - \omega}$$

where:

ω is persistence factor, $0 \leq \omega \leq 1$.

*Recall that we made **four** assumptions about continuing residual income at the end of the short-term period and we will illustrate them as follows:*

Assumption #1: Residual Income Persists at the Current Level Forever

With this assumption, we have $\omega = 1$.

- The residual income in every year after T equals residual income in year T
- The present value of continuing residual income at the end of year $T - 1$ is the present value of a perpetuity.

[LOS 5.h] Explain continuing residual income and justify an estimate of continuing residual income at...

1.

Continuing residual income

PV of continuing RI in year T - 1

$$PV = \frac{RI_T}{1+r-\omega} = \frac{RI_T}{1+r-1} = \frac{RI_T}{r}$$

Assumption #2: Residual Income Drops Immediately to Zero

With this assumption, we have $\omega = 0$.

→ The residual income is expected to drop immediately to zero beginning in year T + 1

PV of continuing RI in year T - 1

$$PV = \frac{RI_T}{1+r-\omega} = \frac{RI_T}{1+r-0} = \frac{RI_T}{1+r}$$

Assumption #3: Residual Income Declines Over Time to Zero

With this assumption, we have $0 < \omega < 1$.

[LOS 5.h] Explain continuing residual income and justify an estimate of continuing residual income at...

1.

Continuing residual income

PV of continuing RI in year $T - 1$

$$PV = \frac{RI_T}{1 + r - \omega} \quad (0 < \omega < 1)$$

Assumption #4: Residual Income Declines to Long-Run Level in Mature Industry

With this assumption, we don't rely on the formula above or ω .

 How to calculate PV of continuing RI in year $T - 1$ with this assumption

Recall from the single-stage residual income model that market value equals book value plus the present value of residual income.

→ At any point in time (T), the present value of future residual income is the difference between market value (P_T) and book value (B_T)

PV of continuing RI in year T

$$PV = P_T - B_T$$

[LOS 5.h] Explain continuing residual income and justify an estimate of continuing residual income at...

1.

Continuing residual income

 *How do we estimate P_T*

Given a forecasted price-to-book ratio and book value at the end of the year T , the value of the stock is:

P_T estimation

$$P_T = B_T \times \text{forecasted P/B ratio}$$

To make this approach consistent with the first three that use the persistence factor equation, we can also calculate the present value of continuing residual income at time $T - 1$:

PV of continuing RI in year $T - 1$

$$PV = \frac{P_T - B_T + RI_T}{1 + r}$$

We will consider some examples in the next slides.

[LOS 5.h] Explain continuing residual income and justify an estimate of continuing residual income at...

1.

Continuing residual income

Example 1: Calculating value with a multistage residual income model (part 1)

Java Metals is expecting an ROE of 15% over each of the next five years. Its current book value is \$5.00 per share, it pays no dividends, and all earnings are reinvested. The required return on equity is 10%. Forecasted earnings in Years 1 through 5 are equal to ROE times beginning book value. Calculate the intrinsic value of the company using a residual income model, assuming that after five years, continuing residual income falls to zero.

Answer:

The following table provides an estimate of the present value of residual income.

Java Metals Residual Income Forecast (*see next slide*)

[LOS 5.h] Explain continuing residual income and justify an estimate of continuing residual income at...

1.

Continuing residual income

Example 1: Calculating value with a multistage residual income model (part 1)

Answer:

Year t	E_t	Ending book value (B_t)	ROE	Equity charge ($r \times B_{t-1}$)	Residual income ($E - r \times B_{t-1}$)
0		\$5.00			
1	\$0.75	5.75	0.15	\$0.50	\$0.25
2	0.86	6.61	0.15	0.57	0.29
3	0.99	7.60	0.15	0.66	0.33
4	1.14	8.74	0.15	0.76	0.38
5	1.31	10.05	0.15	0.87	0.44

[LOS 5.h] Explain continuing residual income and justify an estimate of continuing residual income at...

1.

Continuing residual income

Example 1: Calculating value with a multistage residual income model (part 1)

Answer:

Under the assumption that residual income after five years is zero (i.e., $\omega = 0$), intrinsic value today is:

$$V_0 = \$5.00 + \frac{\$0.25}{1.10} + \frac{\$0.29}{1.10^2} + \frac{\$0.33}{1.10^3} + \frac{\$0.38}{1.10^4} + \frac{\$0.44}{1.10^5}$$

$$= \$6.25.$$

Remember, we can also use your calculator to solve for the answer:



CF0 = 5

C01 = 0.25

C02 = 0.29

C03 = 0.33

C04 = 0.38

C05 = 0.44; I = 10; CPT → NPV = \$6.25.

[LOS 5.h] Explain continuing residual income and justify an estimate of continuing residual income at...

1.

Continuing residual income

Example 2: Calculating value with a multistage residual income model (part 2)

Suppose we change our assumption regarding Java's residual income after five years to assume instead that it remains constant at \$0.44 forever. Calculate the new intrinsic value of Java.

Answer:

The following table provides an estimate of the present value of residual income.

The intrinsic value of Java is higher than the first case because we assume the residual income persists at the same level forever, so $RI_5 = RI_6 = \dots = \$0.44$, and $\omega = 1$. The \$0.44 perpetuity beginning in Year 5 is worth \$4.40 ($\$0.44/0.10$) in Year 4.

The intrinsic value is:

$$\begin{aligned}
 V_0 &= \$5.00 + \frac{\$0.25}{1.10} + \frac{\$0.29}{1.10^2} + \frac{\$0.33}{1.10^3} + \frac{\$0.38 + \$4.40}{1.10^4} \\
 &= \$8.98.
 \end{aligned}$$

[LOS 5.h] Explain continuing residual income and justify an estimate of continuing residual income at...

1.

Continuing residual income

Example 3: Calculating value with a multistage residual income model (part 3)

Now let's make the more realistic assumption that after Year 5, Java's residual income will decay over time to zero with a persistence factor of 0.4. Calculate the new intrinsic value of Java.

Answer:

Residual income begins to decline after Year 5, so the terminal value in Year 4 includes the present value of Year 5 residual income.

$$\text{Terminal value in Year 4} = \frac{RI_T}{1 + r - \omega} = \frac{\$0.44}{1 + 0.10 - 0.40} = \$0.63$$

The intrinsic value today is book value plus the present value of Years 1 through 4 residual income plus the present value of the terminal value in Year 4.

$$\begin{aligned} V_0 &= \$5.00 + \frac{\$0.25}{1.10} + \frac{\$0.29}{1.10^2} + \frac{\$0.33}{1.10^3} + \frac{\$0.38 + \$0.60}{1.10^4} \\ &= \$6.40. \end{aligned}$$

Notice that the more conservative assumption of a lower persistence factor reduces the intrinsic value of the stock because the firm's competitive advantage and economic profits eventually disappear.

[LOS 5.h] Explain continuing residual income and justify an estimate of continuing residual income at...

1.

Continuing residual income

Example 4: Calculating value with a multistage residual income model (part 4)

Suppose instead that at the end of Year 5 we assume that Java's ROE falls to a long run average level and the price-to-book ratio falls to 1.2. Calculate Java's intrinsic value.

Answer:

The book value per share at the end of Year 5 is \$10.05, which means the market price is expected to be $\$10.05 \times 1.2 = \12.06 . The present value of continuing residual income is:

$$\begin{aligned} \text{PV of continuing residual income in Year 4} &= \frac{P_T - B_T + RI_T}{1 + r} = \\ &= \frac{\$12.06 - \$10.05 + \$0.44}{1.10} = \\ &= \$2.23 \end{aligned}$$

Then intrinsic value is:

$$\begin{aligned} V_0 &= \$5.00 + \frac{\$0.25}{1.10} + \frac{\$0.29}{1.10^2} + \frac{\$0.33}{1.10^3} + \frac{\$0.38 + \$2.23}{1.10^4} \\ &= \$7.50. \end{aligned}$$

[LOS 5.i] Compare residual income models to dividend discount and free cash flow models

1.

Comparing RI models to DDM and FCF models

Residual income models

Less sensitive to terminal value estimates

Start with a book value and add to this the present value of the expected stream of residual income to derive the intrinsic value

DDM and FCF models

More sensitive to terminal value estimates

Discount a stream of expected cash flows to derive the intrinsic value

The intrinsic value derived using these models should be identical if the underlying assumptions used to make the necessary forecasts are the same.

If the different models provide dramatically different estimates, the inconsistencies may result from the models' underlying assumptions.

[LOS 5.j] Explain strengths and weaknesses of residual income models and justify the selection of a residual income model to value a company's common stock

1.

Strengths and weaknesses of RI models

Strengths

Terminal value does not dominate the intrinsic value estimate.

Residual income models use accounting data, which is usually easy to find.

The models are applicable to firms that do not pay dividends or that do not have positive expected free cash flows in the short run.

The models are applicable even when cash flows are volatile.

The models focus on economic profitability rather than just on accounting profitability.

Weaknesses

The models rely on accounting data that can be manipulated by management.

Reliance on accounting data requires numerous and significant adjustments.

The models assume that the clean surplus relation holds or that its failure to hold has been properly taken into account.

[LOS 5.j] Explain strengths and weaknesses of residual income models and justify the selection of...

2.

Use cases of RI models

Appropriate circumstances

A firm does not pay dividends, or the stream of payments is too volatile to be sufficiently predictable.

Expected free cash flows are negative for the foreseeable future.

The terminal value forecast is highly uncertain, which makes dividend discount or free cash flow models less useful.

Not appropriate circumstances

There is significant uncertainty concerning the estimates of book value and return on equity.

The clean surplus accounting relation is violated significantly.

[LOS 5.k] Describe accounting issues in applying residual income models

1.

Clean surplus violations

Problems: The clean surplus relationship (i.e., ending book value = beginning book value + net income – dividends) may not hold when items are charged directly to shareholders' equity and do not go through the income statement.

We have to adjust net income to account for these items if they are not expected to reverse in the future.

*There are about **six** items that can bypass the income statement as follows:*

- Foreign currency translation gains and losses that flow directly to retained earnings under the current rate method.
- Certain pension adjustments.
- Gains/losses on certain hedging instruments.
- Changes in revaluation surplus (IFRS only) for long-lived assets.
- Changes in the value of certain liabilities due to changes in the liability's credit risk (IFRS only).
- Changes in the market value of debt and equity securities classified as available-for-sale.

[LOS 5.k] Describe accounting issues in applying residual income models

1.

Clean surplus violations



What is the risk of Clean surplus violations

- Net income is not correct, but book value is still correct.
- ROE forecast will not be accurate if the clean surplus violations are not expected to offset in future years.

Example: Suppose the analyst determines that the clean surplus relation is violated because of the cumulative translation adjustment (CTA) resulting from the application of the current rate method of currency translation. *(See the reading on Multinational Operations if the current rate method doesn't sound familiar!)*

If the CTA tends to reverse over time and is not consistently positive or negative, the ROE can be forecasted without taking into account the CTA.

[LOS 5.k] Describe accounting issues in applying residual income models

2.

Variations from Fair value

Problems: The accrual method of accounting causes many balance sheet items to be reported at book values that are significantly different than their market values.

*There are about **six** common adjustments to the balance sheet necessary to reflect fair value as follows:*

- Operating leases should be capitalized by increasing assets and liabilities by the present value of the expected future operating lease payments. Interest expense should be close to the true cost of debt.
- Special purpose entities (SPEs) whose assets and liabilities are not reflected in the financial statements of the parent company should be consolidated.
- Reserves and allowances should be adjusted.
- Inventory for companies that use LIFO should be adjusted to FIFO by adding the LIFO reserve to inventory and equity, assuming no deferred tax impact.
- The pension asset or liability should be adjusted to reflect the funded status of the plan.
- Deferred tax liabilities should be eliminated and reported as equity if the liability is not expected to reverse.

[LOS 5.k] Describe accounting issues in applying residual income models

3.

Intangible asset effects on Book value

*There are about **two** intangible assets require special attention: (1) intangibles recognized at acquisition and (2) R&D expenditures.*

Problems: Recognition of an identifiable intangible asset (such as a license) in the group accounts that was not previously recorded in the investee company balance sheet creates a distortion in valuation under a residual income model.

It reduces the combined ROE → lower valuation of the combined entity compared to the sum of the values of individual entities prior to acquisition.

To remove this distortion, the **amortization** of intangibles capitalized during acquisition should be **removed** prior to computing the ROE used for residual income valuation.

The suggested analytical treatment of **R&D expenditures** is less definitive, but we can make the general statement that the ROE estimate for a mature company should reflect the long-term productivity of the company's R&D expenditures:

- Productive R&D expenditures increase ROE and residual income, and
- Unproductive expenditures reduce ROE and residual income.

[LOS 5.k] Describe accounting issues in applying residual income models

4.

Nonrecurring items

Problems: Companies often report non-recurring charges as part of earnings, which can lead to overestimates and underestimates of future residual earnings if no adjustments are made.

Nonrecurring items should not be included in residual income forecasts because they represent items that are not expected to continue in the future.

*There are about **five** items that may need adjustment in measuring recurring earnings include:*

- Discontinued operations
- accounting changes
- Unusual items
- Extraordinary items (applicable under US GAAP but not under IFRS)
- Restructuring charges.

Firms may adopt other types of aggressive accounting practices that overstate the book value of assets and earnings by, for example, accelerating revenues to the current period or deferring expenses to a later period.

[LOS 5.k] Describe accounting issues in applying residual income models

5.

International accounting differences

Problems: Residual income models, which are based on accrual accounting information, may not be as useful in valuing foreign firms because of differences in national accounting standards.

*There are **three** primary considerations in applying a residual income model internationally:*

- The availability of reliable earnings forecasts
- Systematic violations of the clean surplus assumption
- “Poor quality” accounting rules that result in delayed recognition of value changes

Analysts should expect the model to work best in situations in which earnings forecasts are available, clean surplus violations are limited, and accounting rules do not result in delayed recognition.

MODULE 6 : Private Company Valuation

LEARNING OUTCOMES

- 6.a.** Contrast important public and private company features for valuation purposes.
- 6.b.** Describe uses of private business valuation and explain key areas of focus for financial analysts.
- 6.c.** Explain cash flow estimation issues related to private companies and adjustments required to estimate normalized earnings.
- 6.d.** Explain factors that require adjustment when estimating the discount rate for private companies.
- 6.e.** Compare models used to estimate the required rate of return to private company equity (for example, the CAPM, the expanded CAPM, and the buildup approach).
- 6.f.** Explain and evaluate the effects on private company valuations of discounts and premiums based on control and marketability.
- 6.g.** Explain the income, market, and asset-based approaches to private company valuation and factors relevant to the selection of each approach.

LEARNING OUTCOMES

- 6.h.** Calculate the value of a private company using income-based methods.
- 6.i.** Calculate the value of a private company using market-based methods and describe the advantages and disadvantages of each method.

[LOS 6.a] Contrast important public and private company features for valuation purposes

In previous modules, we have focused on valuation of publicly held company with periodic audited financial statements and an observable market-based share price. But this is not always the case. There are private companies whose shares are not listed on public markets.

Example: Form of private companies



Sole proprietorship: business that can be owned and controlled by an individual, a company or a limited liability partnership. It is not a separate legal entity from its owner.



Privately held corporation: a firm whose ownership shares or interest are not publicly traded. Private companies may issue stock and have shareholders, but their shares are not listed on public markets.



Formerly public company: a company whose stock used to be publicly traded, but has been taken private in buyouts or other transactions.

The characteristics that distinguish private and public companies can be delineated into *company-specific* and *stock-specific* factors (introduced next).

[LOS 6.a] Contrast important public and private company features for valuation purposes

Public Versus Private Company Features

Public company

Mature company

Liquid shares

Owner/ Manager
separation

Standardized
disclosures

Greater
transparency

Private company

Early/Late Life
Cycle Phase

Illiquidity

Smaller Size

Concentrated
Control

Concentrated
Ownership

Sale Restrictions

Limited
Disclosures

**2. Stock specific
factors**

Owner/Manager
Overlap

**1. Company
specific factors**

[LOS 6.a] Contrast important public and private company features for valuation purposes

1.

Company-specific factors

	Private Company	Public Company
Stage of life cycle	Usually in early or late stages of business cycle.	More mature than private companies.
Size	• Have less capital, fewer assets and fewer employees. • Lack of access to public equity markets can constrain growth.	• Operate in larger scale. Have access to public equity markets. • Have more regulatory burdens which can outweigh the benefit of greater access to funds.
Quality and depth of management	Cannot attract as many qualified applicants.	Able to draw in many quality candidates.

[LOS 6.a] Contrast important public and private company features for valuation purposes

1.

Company-specific factors

	Private Company	Public Company
Management/ shareholder overlap	In most private firms, management has a substantial ownership position, reducing principal-agent conflict.	Have separated roles of BODs/BOMs to secure the interests of minor shareholders.
Short-term investors	Managers are long-term holders of significant equity interests.	Shares are traded on exchanges, allow short-term investors with short-term goals join the board.
Quality of financial and other information	Less requirements and regulations for financial reporting.	Public firms are required to make timely, in-depth financial disclosures.
Tax consideration	More concern due to the effect on private equity shareholders/managers.	Less concerned with tax in comparison with private company.

[LOS 6.a] Contrast important public and private company features for valuation purposes

2.

Stock-specific factors

	Private Company	Public Company
Liquidity on shares	Have fewer potential owners and is less liquid than publicly traded equity. → A liquidity discount is often applied in valuing privately held shares.	Have listed shares and more shareholders, their shares are traded more frequently.
Restrictions on marketability	Often have agreements that prevent shareholders from selling, reducing the marketability of shares.	Public listed shares don't usually have any restriction on sales of shares.
Concentration of control	The control of private firms is usually concentrated in the hands of a few shareholders.	• More separated roles of owners and managers. • Minority interest have more protections in public firms.

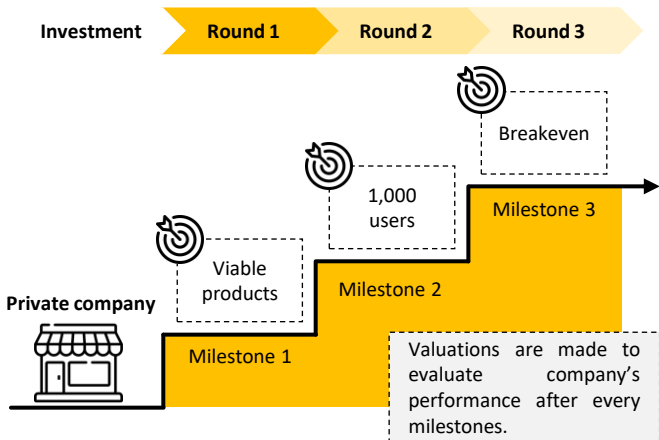
[LOS 6.b] Describe uses of private business valuation and explain key areas of focus for financial analysts.

1.

Transaction-Related Valuations

Venture capital financing

A type of funding by venture capital. It is private equity capital that can be provided at various funding rounds, after the company achieved a *milestone*.



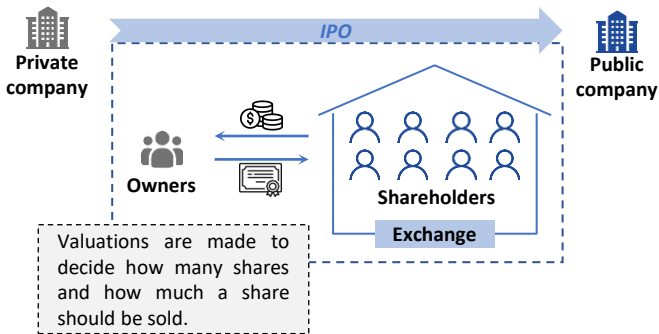
[LOS 6.b] Describe uses of private business valuation and explain key areas of focus for financial analysts.

1.

Transaction-Related Valuations

Initial public offering (IPO)

The process of a private firm offering its shares to the public by a stock issuance **for the first time**. After an IPO, firm will have access to public equity market through stock exchanges.



IPOs are often conducted under the following circumstances:

- An early-stage firm wants to attract public equity investment.
- A new firm from a line of business from an existing public company.
- A firm which was previously held by the public returns to public markets following a restructuring phase under private ownership.

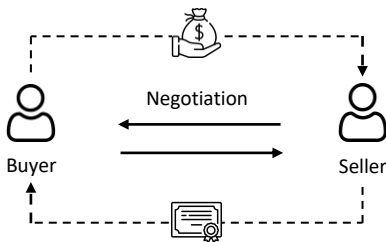
[LOS 6.b] Describe uses of private business valuation and explain key areas of focus for financial analysts.

1.

Transaction-Related Valuations

Sale in an acquisition

A sale of a development-stage or mature private firm to generate liquidity for the shareholders. It can be seen as a “cash-out” for business owners.



Valuations are usually performed by both the firm and the buyer to support the negotiation of both parties.

[LOS 6.b] Describe uses of private business valuation and explain key areas of focus for financial analysts.

1.

Transaction-Related Valuations

Bankruptcy proceedings

For firms in bankruptcy, accurate valuation can help determine whether the firm should be liquidated or reorganized.



Company facing bankruptcy



Valuation



Reorganization



Liquidation

[LOS 6.b] Describe uses of private business valuation and explain key areas of focus for financial analysts.

1.

Transaction-Related Valuations

**Performance-
based
managerial
compensation**

If a firm compensates employees with stock options, grants of restricted stock, or employee stock ownership plans, accurate valuation is necessary for both accounting and tax purposes.

Debt financing

Both issuers and lenders of private companies can perform a valuation to determine a firm's ability to repay existing debt from current operating cash flows.

[LOS 6.b] Describe uses of private business valuation and explain key areas of focus for financial analysts.

2.

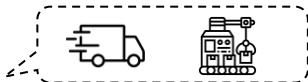
Compliance-Related Valuations

Financial reporting

Components and divisions of public companies require ongoing valuations for performance reporting and measurement purposes.



Public company making its financial reports



Unlisted business units

Valuation of BUs for financial reporting purposes of the parent company

Tax purposes

- **At the firm level**, transfer pricing, property taxes, and corporate restructuring may necessitate valuations.
- **For individual equity owners**, estate and gift tax issues may necessitate valuations.

[LOS 6.b] Describe uses of private business valuation and explain key areas of focus for financial analysts.

3.

Litigation-Related Valuations

Litigation-related valuations



Shareholder
suits



Damage
claims



Insurance
claims



Divorce
settlements

4.

Performers of types of valuation

Because the valuation methods for transactions, compliance, and litigation are often quite different, most appraisers specialize in a single area.

Transaction-
related

Usually performed by investment bankers;

Compliance-
related

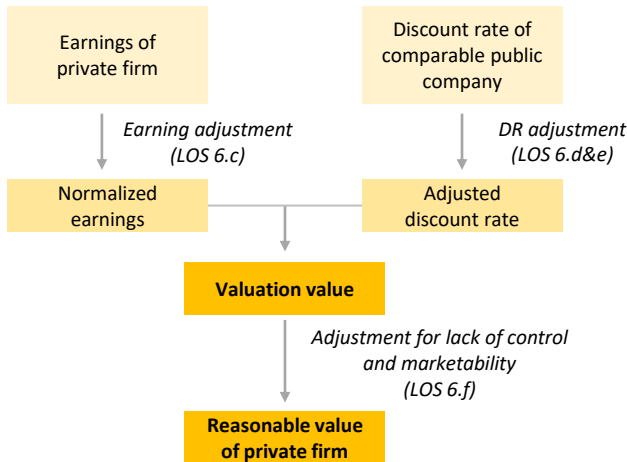
Usually performed by those with accounting or tax knowledge;

Litigation-
related

Usually performed by those comfortable with a legal setting and specific jurisdictions.

[Get ready] Describe the valuation process of a private company

Financial reportings in private companies does not undergone strict regulations as in public firms, and many other characteristics of private firms and discussed earlier can affect its financial statements. Figures from the financial reports used as inputs for valuations, therefore, can lead to misassumption and incorrectly interpret the company's value.



[LOS 6.c] Explain cash flow estimation issues related to private companies and adjustments required to estimate normalized earnings.

1.

Estimating Normalized Earnings

Normalized Earnings

Normalized earnings are earnings that are adjusted to remove the effect of unusual or one-time revenue or expenses. Beside one-time events, private equity valuation faces **ongoing distortions** that also need to be put into consideration.

What's the problem with ongoing distortions in private companies?

Ongoing distortions result from revenues or expenses arising from related-party transactions.

Example:

- Transactions occurring between a private company and its controlling owners and are often related to compensation or non-operating assets.
- Transactions occurring between companies related to the owners' of private firms.

Because these transactions do not take place in a fair market, they are not recorded at the fair market value. Records of such transactions **are not a valid base to perform valuation**.

[LOS 6.c] Explain cash flow estimation issues related to private companies and adjustments required to estimate normalized earnings.

1.

Estimating Normalized Earnings

Parts of the Profit and Loss report that need adjustments for ongoing distortions

Revenue

Market pricing and terms for sales between affiliates

COGS

Related party compensation, real estate use or other items

**Depreciation &
Amortization**

Distortions/Adjustments related to asset value, ownership

Taxes

Adjust taxes based upon adjusted taxable income

**Dividends &
Distributions**

Adjust for differences between private owner managed and public company shareholder distributions

[LOS 6.c] Explain cash flow estimation issues related to private companies and adjustments required to estimate normalized earnings.

1.

Estimating Normalized Earnings

Parts of the Profit and Loss report that need adjustments for ongoing distortions

Some common adjustments

Circumstances

Adjustments

Above-market or under-market compensation for managers

Adjust compensation to market based level.

The company is using owned property in its business operations

Add a market rental charge for the use of the real estate to the expenses of the company.

Real estates that are not used for core operation

Remove the income and expense related to that real estate.

[LOS 6.c] Explain cash flow estimation issues related to private companies and adjustments required to estimate normalized earnings.

1.

Estimating Normalized Earnings

Example: Estimating Normalized Earnings

Tim Groh is the principal shareholder, CEO, and founder of Arbutus Generators (Arbutus). Arbutus reports the three following annual figures:

- Groh's compensation of \$2.5 million is included in the firm's selling, general, and administrative (SG&A) expenses.
- Arbutus leases a warehouse for \$100,000 a year from one of its largest suppliers.
- Arbutus owns a vacant office building with reported SG&A expenses of \$150,000 and \$25,000 of depreciation expense.

An analyst determines that a market-based compensation for Groh's position is \$1 million annually and that the office building is not needed for core operations. The market lease rate of the warehouse is \$130,000 per year.

Based on the three annual figures just listed, what adjustments should the analyst make to Arbutus's reported income to estimate normalized earnings (earnings), assuming the firm will be acquired?

[LOS 6.c] Explain cash flow estimation issues related to private companies and adjustments required to estimate normalized earnings.

1.**Estimating Normalized Earnings****Example: Estimating Normalized Earnings****Answer:**

- a. Because the market rate is \$1.5 million less than Groh's actual compensation, SG&A expenses should be reduced by \$1.5 million to reflect a normalized compensation expense.
- b. Because the market lease rate is \$30,000 higher than the reported lease expense, SG&A expenses should be increased by \$30,000 to reflect a normalized lease rate.
- c. Because the office building is noncore, SG&A expenses should be reduced by \$150,000, and the depreciation expense should be reduced by \$25,000.

[LOS 6.c] Explain cash flow estimation issues related to private companies and adjustments required to estimate normalized earnings.

2.

Estimating Cash Flow



FCFF

Free cash flow to the firm (FCFF): the cash flow generated by the firm's operations that is in excess of the capital investment required to sustain the firm's current productive capacity.



FCFE

Free cash flow to equity (FCFE): the cash available to stockholders after funding capital requirements and expenses associated with debt financing. This is the cash flow used to directly value equity.

- As noted previously, the valuation of equity depends on the definition of value used. Also, controlling and noncontrolling equity interests will have quite different values.
→ These differences should be accounted for in cash flow estimates and assumptions.
- When there is significant uncertainty about a private company's future operations, the analyst should examine several scenarios when estimating future cash flows.

[LOS 6.c] Explain cash flow estimation issues related to private companies and adjustments required to estimate normalized earnings.

2.

Estimating Cash Flow

Example: Estimating Cash Flow

An analyst has normalized the earnings and expenses for a private firm under consideration as an acquisition. Because the capital structure is nonoptimal, the analyst assumes that the capital structure will be changed if the firm is acquired and will use the FCFF approach to value the firm. The following assumptions are used to create a pro forma income statement and to estimate FCFF.

Current revenues	\$20,000,000
Revenue growth	4%
Gross profit margin	30%
Depreciation expense as a percentage of sales	2%
Working capital as a percentage of sales	10%
SG&A	\$2,200,000
Tax rate	30%

Additionally, capital expenditures will cover depreciation plus 6% of the firm's incremental revenues.

Create a pro forma income statement and estimate FCFF.

[LOS 6.c] Explain cash flow estimation issues related to private companies and adjustments required to estimate normalized earnings.

2.

Estimating Cash Flow

Example: Estimating Cash Flow

Pro Forma Income Statement

Revenue	\$20,800,000
Cost of goods sold	<u>\$14,560,000</u>
Gross profit	\$6,240,000
SG&A	\$2,200,000
Pro forma EBITDA	\$4,040,000
Depreciation and amortization	<u>\$416,000</u>
Pro forma EBIT	\$3,624,000
Pro forma taxes on EBIT	<u>\$1,087,200</u>
Operating income after tax	\$2,536,800
<i>Adjustments to obtain FCFF</i>	
Plus: Depreciation and amortization	\$416,000
Minus: Capital expenditures	\$464,000
Minus: Working capital	<u>\$80,000</u>
FCFF	\$2,408,800

[LOS 6.c] Explain cash flow estimation issues related to private companies and adjustments required to estimate normalized earnings.

2.

Estimating Cash Flow

Example: Estimating Cash Flow

Pro Forma Income Statement

Revenue	Current revenues times the growth rate: $\$20,000,000 \times 1.04$
Cost of goods sold	Revenues times one minus the gross profit margin: $\$20,800,000 \times (1 - 0.30)$
Gross profit	Revenues times the gross profit margin: $\$20,800,000 \times 0.30$
SG&A expense	Given in the question
Pro forma EBITDA	Gross profit minus SG&A expenses: $\$6,240,000 - \$2,200,000$
Depreciation and amortization	Revenues times the given depreciation expense: $\$20,800,000 \times 0.02$
Pro forma EBIT	EBITDA minus depreciation and amortization: $\$4,040,000 - \$416,000$
Pro forma taxes on EBIT	EBIT time tax rate: $\$3,624,000 \times 0.30$
Operating income after tax	EBIT minus taxes: $\$3,624,000 - \$1,087,200$

[LOS 6.c] Explain cash flow estimation issues related to private companies and adjustments required to estimate normalized earnings.

2.
Estimating Cash Flow
Example: Estimating Cash Flow

Adjustments to obtain FCFF	Explanation
Plus: Depreciation and amortization	Add back noncash charges from those listed previously
Minus: Capital expenditures	Expenditures cover depreciation and increase with revenues: $\$416,000 + [0.06 \times (\$20,800,000 - \$20,000,000)]$
Minus: Working capital	Working capital will increase as revenues increase: $0.10 \times (\$20,800,000 - \$20,000,000)$
FCFF	Operating income net of the adjustments listed previously

[LOS 6.d] Explain factors that require adjustment when estimating the discount rate for private companies.

Adjust discount rate



Challenging

Refer to LOS 6.d



Model to estimate

Refer to LOS 6.e

1. CAPM

2. Expanded
CAPM

3. The build-up
method

[LOS 6.d] Explain factors that require adjustment when estimating the discount rate for private companies.

Several factors make estimating a rate at which to discount a private company's expected future cash flows challenging.

Size premiums

Premiums are often added to the discount rates used for small private companies. *(The smaller the riskier)*

Relative debt availability and cost of debt

Private companies may have less access to debt financing than a similar public company.
 → *They rely more on equity financing*
 → *Increase WACC*

Acquirer versus target

Acquirers can incorrectly use their own (lower) cost of capital. The appropriate cost of capital for the target is usually higher

Projection risk

The lower availability of information about private firms may make it appropriate for an analyst to apply a higher discount rate.

Life cycle stage

Ranges of discount rates can be specified for the various life cycle stages, it may be difficult to accurately classify the stage a firm is in.

[LOS 6.e] Compare models used to estimate the required rate of return to private company equity (for example, the CAPM, the expanded CAPM, and the build-up approach).

1.

CAPM

We have the Capital Asset Pricing Model (CAPM) formula

$$R_i = R_f + \beta \times ERP$$

Investments in private companies may be large, unique, and undiversified, the company is usually subjected to a higher risk premium than suggested by β .

To use β in valuation for private firms, adjustments to leverage is needed

Step 1

Unlever the comparable public company beta using its tax rate and debt-to-equity ratio

$$\beta_{\text{unlevered}} = \frac{\beta_{\text{public}}}{[1 + (1 - t)(\frac{D}{E})]}$$

Step 2

Adjust this unlevered beta using the subject private company's tax rate (t^*) and leverage (D/E)^{*}

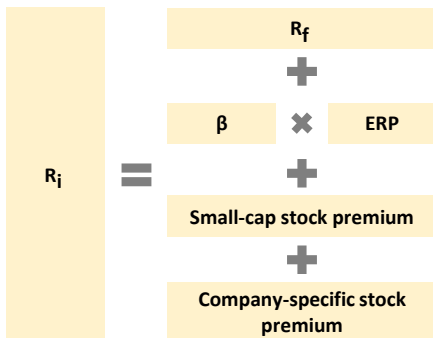
$$\beta_{\text{private company}} = \beta_{\text{unlevered}} [1 + (1 - t^*)(\frac{D}{E})^*]$$

[LOS 6.e] Compare models used to estimate the required rate of return to private company equity...

2.

Expanded CAPM

Expanded CAPM includes additional premiums for size and firm-specific (unsystematic) risk.

$$R_i = R_f + \beta \times \text{ERP} + \text{Small-cap stock premium} + \text{Company-specific stock premium}$$


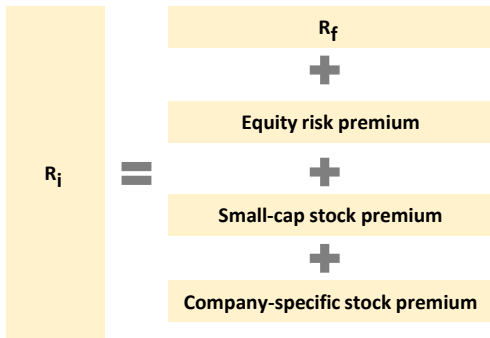
The diagram illustrates the Expanded CAPM formula. On the left, a vertical yellow box contains the variable R_i . To its right is an equals sign. Further right, the formula is built vertically: a yellow box with R_f is followed by a plus sign, then a yellow box with β followed by a multiplication sign and a yellow box with ERP, then another plus sign, then a yellow box with 'Small-cap stock premium', then a third plus sign, and finally a yellow box with 'Company-specific stock premium'.

[LOS 6.e] Compare models used to estimate the required rate of return to private company equity...

3.

The build-up method

Used when it is not possible to find comparable public firms for beta estimation. Premiums then are added for small size, industry factors and company-specific factors.



[LOS 6.e] Compare models used to estimate the required rate of return to private company equity...

3.

The build-up method

Example:

An analyst is examining a private firm being considered for acquisition and determines the following:

- The current capital structure is nonoptimal because the owner avoids using debt.
- A small stock premium and company-specific risk premium are appropriate because the private firm is much smaller and much less diversified than the public firms that beta is estimated from.
- An industry risk premium is needed to account for the additional risk in this industry compared to the broad market.

The relevant figures are as follows.

Continued in next slide

[LOS 6.e] Compare models used to estimate the required rate of return to private company equity...

3.

The build-up method

Risk-free rate	3.6%
Equity risk premium	6.0%
Beta	1.3
Small stock premium	3.0%
Company-specific risk premium	2.0%
Industry risk premium	1.0%
Pretax cost of debt	9.0%
Debt/total cap for public firms in industry	30%
Optimal debt/total cap	12%
Current debt/total cap	3%
Tax rate	30%

Questions:

- Calculate the required return on equity using the CAPM, the expanded CAPM, and the build-up method.
- Calculate the WACC using the current capital structure and the optimal capital structure, assuming a cost of equity of 16%.
- Comment on the appropriate capital structure weights.

[LOS 6.e] Compare models used to estimate the required rate of return to private company equity...

Answer:

- a. The required return on equity using the CAPM is as follows: required return = $R_f + (\beta \times ERP) = 3.6\% + 1.3(6\%) = 11.4\%$.

Using the expanded CAPM, a small stock premium and company-specific risk premium are added: $11.4\% + 3\% + 2\% = 16.4\%$.

Using the build-up method, beta is omitted (but an industry risk premium is added): $3.6\% + 6\% + 1\% + 3\% + 2\% = 15.6\%$.

- b. The WACC, using current capital structure, is as follows:

$$WACC = (w_e \times r_e) + [w_d \times r_d \times (1 - \text{tax rate})]$$

$$= (0.97 \times 16\%) + [0.03 \times 9\% \times (1 - 30\%)]$$

$$= 15.7\%$$

The WACC, using the optimal capital structure, is as follows:

$$= (0.88 \times 16\%) + [0.12 \times 9\% \times (1 - 30\%)]$$

$$= 14.8\%$$

- c. The current capital structure reflects the current owner's conservative use of debt. The optimal capital structure can be determined through discussions with financiers. The optimal capital structure should be used to calculate the (lower) WACC for the acquisition, reflecting the higher level of debt the firm can support. The capital structure for public firms in the same industry should not be used because public firms are likely to have better access to debt financing. A public firm could likely take on more (less expensive compared to equity) debt than a private company. For this reason, a private firm will likely have a greater WACC than a public firm in the same industry.

[LOS 6.f] Explain and evaluate the effects on private company valuations of discounts and premiums based on control and marketability.

1.

Strategic and Nonstrategic Buyers

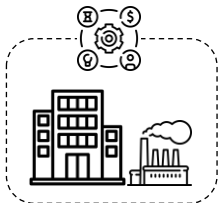
Strategic transaction

- The acquired assets have synergies with acquirer's existing assets.
- Valuation of the firm is based in part on the perceived synergies with the acquirer's other assets.

Financial transaction

- The acquired assets have no synergies.
- Valuation of the firm is conducted independently of the acquirer's assets.

Estimating normalized earnings for a strategic transaction



Perceived synergy

Increase revenues
Reduce cost

Adjustments result in higher
valuation value

[LOS 6.f] Explain and evaluate the effects on private company valuations of discounts and premiums based on control and marketability.

1.

Strategic and Nonstrategic Buyers

Example: Incorporating synergies

An analyst is valuing a firm with two different potential buyers. Buyer A is a firm in the same industry as the target firm, which expects to reduce costs at the target firm by eliminating redundancies. Buyer B is a firm in an unrelated industry.

Calculate the normalized EBITDA for each buyer given the following information

Reported EBITDA	\$4,800,000
Current executive compensation	\$900,000
Market-based executive compensation	\$600,000
Current SG&A expenses	\$8,000,000
SG&A expenses after synergistic saving	\$7,600,000

Answer:

- Both the strategic (Buyer A) and nonstrategic (Buyer B) buyers will attempt to reduce executive compensation to market levels. So, the adjustment for both buyers to generate normalized EBITDA is $\$4,800,000 + (\$900,000 - \$600,000) = \$5,100,000$.
- Only Buyer A will be able to realize synergistic savings of $(\$8,000,000 - \$7,600,000) = \$400,000$. So, normalized EBITDA for Buyer A is $\$5,500,000$; for Buyer B, it is $\$5,100,000$.

[LOS 6.f] Explain and evaluate the effects on private company valuations of discounts and premiums based on control and marketability.

2.

Discount for Lack of Control (DLOC)

Discount for Lack of Control

Minority shareholders are at a disadvantage because they have less power to influence the company. Therefore, adjustment for lack of control is required.



Controlling
shareholders with
too many perks

$$\text{DLOC} = 1 - \left[\frac{1}{1 + \text{control premium}} \right]$$

Adjustment for lack
of control



Valuation
conducted at the
perspective of a
minor shareholder

Depending on the estimated and subject cash flows were based on controlling or noncontrolling interest, the analyst should decide if the adjustment is required or not.

[LOS 6.f] Explain and evaluate the effects on private company valuations of discounts and premiums based on control and marketability.

3.

Discount for Lack of Marketability (DLOM)

**Discount for
Lack of
Marketability
(Liquidity)**

The adjustment in valuation applied for firms that cannot be easily sold.

It is often the case that if a DLOC is applied, a DLOM will also be appropriate.

Factors affect firm's marketability will also impact DLOM

Impending IPO or firm sale



Increase
marketability



Decrease DLOM

Contractual restrictions
on selling stock



Decrease
marketability



Increase DLOM

Larger pool of buyers



Increase
marketability



Decrease DLOM

Greater ownership
concentration



Increase
marketability



Increase DLOM

[LOS 6.f] Explain and evaluate the effects on private company valuations of discounts and premiums based on control and marketability.

3.

Discount for Lack of Marketability (DLOM)

Three methods to estimate DLOM

Compare the price of the **restricted shares** to the price of the **publicly traded shares**.

Compare the price of **pre-IPO shares** to that of **post-IPO shares**.

There are more factors affect price of shares after an IPO than marketability (eg: lower risk, more certain cash flows), so the estimated DLOM may not be correct

Estimates the DLOM as **the value of an at-the-money put option** of a comparable public company, **divided by its stock price**.

Advantage: the estimated risk of the private company can be factored into the option price.

Drawback: a put option indicates a certain selling price—not actual liquidity

Note: The three methods only provide a basis for calculating DLOM, the magnitude of the DLOM applied to a company will vary by analyst.

[LOS 6.f] Explain and evaluate the effects on private company valuations of discounts and premiums based on control and marketability.

3.**Discount for Lack of Marketability (DLOM)**

The DLOC and DLOM are applied sequentially, therefore they are multiplicative, not additive.

If the DLOC is 20%, and the DLOM is 13%, the total discount is **NOT** calculated as $20\% + 13\% = 33\%$, but as follows:

Total discount = $1 - [(1 - \text{DLOC})(1 - \text{DLOM})] = 1 - [(1 - 0.20)(1 - 0.13)] = 30.4\%$

[LOS 6.f] Explain and evaluate the effects on private company valuations of discounts and premiums based on control and marketability.

Example: Calculating the value of a minority interest

A minority shareholder holds 15% of a private firm's equity, and the CEO holds the other 85%. Suppose that there are two possible scenarios.

In Scenario 1, the CEO is likely to sell the firm soon. In this case, the valuation discounts will be small. A DLOM of 5% will be applied. A DLOC will not be applied, under the assumption that all selling shareholders will receive the same price. The value of the firm's equity is estimated at \$10 million.

In Scenario 2, the CEO has no plans to sell the firm, and the minority shareholder cannot sell their interest easily. A DLOM of 20% will be applied. A DLOC will be estimated by using reported earnings instead of normalized earnings to provide an estimated firm equity value of \$9 million. The \$9 million value assumes that certain firm inefficiencies (e.g., above-market compensation for the owner) cannot be corrected without a sale of the firm.

Given these figures, calculate the value of the minority shareholder's equity interest under both scenarios

Answer:

The following provides the calculations under each scenario.

[LOS 6.f] Explain and evaluate the effects on private company valuations of discounts and premiums based on...

Scenario 1: Assuming sale is likely

Firm's equity value	\$10,000,000
Minority interest	15%
Value of minority interest without discounts	%1,500,000
Minus DLOC of 0%	0
Value of interest if marketable	\$1,500,000
Minus DLOM of 5%	\$75,000
Value of minor interest	\$1,425,000

Scenario 2: Assuming sale is unlikely

Firm's equity value	\$9,000,000
Minority interest	15%
Value of minority interest without discounts	\$1,350,000
Minus DLOC of 0%	0
Value of interest if marketable	\$1,350,000
Minus DLOM of 5%	\$270,000
Value of minor interest	\$1,080,000

The smaller value of the minority interest in Scenario 2 is due to the higher DLOM and the DLOC (as selected in the lower firm equity value of \$9 million)

[LOS 6.g] Explain the income, market, and asset-based approaches to private company valuation and factors relevant to the selection of each approach

Three major approaches to private company valuation:

Income approach

Values a firm as the present value of its expected future income.

Market approach

Values a firm using multiples from recent sales of comparable assets.

Asset-based approach

Values a firm as the value of its assets minus its liabilities

Factors to consider to select the appropriate valuation method



Business cycle

In early stages, asset-based is most appropriate while in mature stages, income approach is more appropriate.



Company size

In relative methods, comparable companies should have the similar size.



Types of assets

A valuation based on income or earnings multiples would not attach a value to nonoperating assets.

[LOS 6.h] Calculate the value of a private company using income-based methods

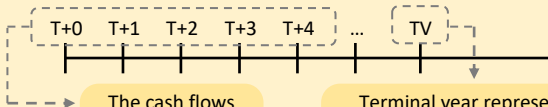
1.

The Free Cash Flow Method

The Free Cash Flow method

A series of discrete cash flows are forecastED, plus a terminal value that reflects the value of the business as a going concern at some future date

The terminal value can be calculated using a *constant growth model*, *price multiple approach* or the *capitalized cash flow method* (introduced next).



The cash flows are forecast with company characteristics.

Terminal year represent the point of time at which growth is expected to level off and remain constant.

[LOS 6.h] Calculate the value of a private company using income-based methods

2.

The Capitalized Cash Flow Method

The Capitalized Cash Flow Method

A single measure of economic benefit is divided by a capitalization rate - equal to the required rate of return minus a growth rate - to arrive at firm value.

The capitalized cash flow method is most often used for small private companies where stable growth is a reasonable assumption.

Valuing the firm as a whole using the CCM

$$\text{Value of the firm} = \frac{\text{FCFF}_1}{\text{WACC} - g}$$

Where:

FCFF_1 = expected free cash flow to the firm over the next year

WACC = weighted average cost of capital (assuming a constant capital structure)

g = sustainable growth rate in free cash flows

Alternative formulas to estimate value of firm/equity

$$b = g / \text{WACC}$$

$$\text{Value of the firm} = \frac{\text{EBIT}_1(1 - T)(1 - b)}{\text{WACC} - g}$$

$$\text{Value of equity} = \frac{\text{FCFE}_1}{r - g}$$

[LOS 6.h] Calculate the value of a private company using income-based methods

2.

The Capitalized Cash Flow Method

Example: Calculating firm value using the capitalized cash flow method

Given the following figures, calculate the value of the firm and of the firm's equity using the CCM.

FCFF in one year	\$12,100,000
Growth rate of FCFF	4.0%
WACC	15.0%
Market value of debt	\$4,000,000

Answer:

Step 1: Calculate the value of the firm.

Using the FCFF formula:

$$\text{value of firm} = \text{FCFF1} / (r - g) = (\$12,100,000) / (0.15 - 0.04) = \$110,000,000$$

Step 2: Calculate the value of the equity.

Subtract the debt value from firm value:

$$\text{value of equity} = \text{firm value} - \text{debt value} = \$110,000,000 - \$4,000,000 = \$106,000,000$$

Note that the capitalization rate in this example is $15\% - 4\% = 11\%$. The WACC will be greater when more (relatively expensive) equity and less (cheap) debt are used, resulting in lower estimates of firm and equity values.

[LOS 6.h] Calculate the value of a private company using income-based methods

3.

The Excess Earnings Method

The Excess Earnings Method

Step 1: Calculate the required return for working capital and fixed assets.

Step 2: Calculate the excess earnings.

Step 3: Value the intangible assets.

Step 4: Sum the asset values to arrive at the total firm value.

The excess earnings method (EEM) is employed only infrequently, but it can be useful for small firms when their intangible assets are significant. However, the required return for working capital and fixed assets is subject to estimation error.

[LOS 6.h] Calculate the value of a private company using income-based methods

3. The Excess Earnings Method

Example: Calculating firm value using the EEM

Given the following figures, calculate the value of the firm using the EEM.

Working capital	\$300,000
Fixed assets	\$1,000,000
Normalized earnings (year just ended)	\$130,000
Required return for working capital	6%
Required return for fixed assets	10%
Growth rate of residual income	5%
Discount rate for intangible assets	14%

Answer:

Step 1: Calculate the required return for working capital and fixed assets. Based on the required rates of return for working capital and fixed assets, these are required earnings:

Required return on working capital: $\$300,000 \times 6\% = \$18,000$

Required return fixed assets: $\$1,000,000 \times 10\% = \$100,000$

Step 2: Calculate the excess earnings. excess earnings = $\$130,000 - \$18,000 - \$100,000 = \$12,000$

[LOS 6.h] Calculate the value of a private company using income-based methods

3.

The Excess Earnings Method

Example: Calculating firm value using the EEM

Step 3: Value the intangible assets.

Using the formula for a growing perpetuity, the discount rate for intangible assets, and the growth rate for excess earnings, this is the value:

$$\text{Value of intangible assets} = (\$12,000 \times 1.05) / (0.14 - 0.05) = \$140,000$$

Step 4: Sum the asset values to arrive at the total firm value.

$$\begin{aligned} \text{Firm value} &= \text{working capital} + \text{ixed assets} + \text{intangible assets} \\ &= \$300,000 + \$1,000,000 + \$140,000 = \$1,440,000 \end{aligned}$$

[LOS 6.i] Calculate the value of a private company using market-based methods and describe the advantages and disadvantages of each method

1.

Guideline Public Company Method

Guideline Public Company Method

Uses enterprise value multiples for public companies, with adjustments to the multiples to account for differences in size, leverage, and life cycle stage between the subject firm and the comparables.

The control premium (i.e., the value of control) should be estimated. The control premium equals the difference between the pro rata value of a controlling interest and the pro rata value of a noncontrolling interest.

[LOS 6.i] Calculate the value of a private company using market-based methods and describe the advantages and disadvantages of...

1.

Guideline Public Company Method

Factors to consider when estimating a control premium

1. **Transaction type.** Recall that a transaction may be either strategic or financial.
2. **Industry conditions.** Adding premium in an active acquisition market may overstate the appropriate premium for control.
3. **Type of consideration.** Some acquisitions are settled with inflated shares and will be overstated if shares were traded at inflated price.
4. **Reasonableness.** The use of control premiums and price multiples can sometimes compound into significant deviations from historical pricing.
5. **Multiple industries.** When using a comparables-based approach, a private company operating in multiple industries should be assigned a weighted average multiple based on the proportion of revenues generated in each industry:

$$\text{weighted average EV multiple} = \sum W_i \text{ multiple}$$

where: i = industry and W_i = weight of Industry i in revenues of the subject company
6. **Other factors.** Additional considerations include differences in size, country, tax status, and leverage.

[LOS 6.i] Calculate the value of a private company using market-based methods and describe the advantages and disadvantages of...

1.

Guideline Public Company Method

Example: Valuation using the guideline public company method

An analyst, Natalie Hoskins, is valuing a private firm, Rensselaer Components (Rensselaer), using the GPCM and EV to EBITDA multiples. Hoskins has gathered data for comparable public firms indicating an average multiple of 1.18. Rensselaer operates primarily in the computer component manufacturing industry (70% of revenues), but it derives 20% and 10% of its revenues from the defense and telecom industries, respectively. The average EV/EBITDA multiple for defense and telecom industries is 1.02 and 2.45, respectively. Rensselaer's normalized EBITDA is \$12.80 million.

- Determine an appropriate enterprise value to EBITDA multiple to be used to value Rensselaer.
- Estimate the enterprise value for Rensselaer using EV/EBITDA.

Answer:

- $\text{EV/EBITDA multiple} = (0.70 \times 1.18) + (0.20 \times 1.02) + (0.10 \times 2.45) = 1.28.$
- $\text{Rensselaer's estimated enterprise value} = \text{EBITDA} \times \text{EV/EBITDA} = 12.80 \times 1.28 = \$16.38 \text{ million}.$

[LOS 6.i] Calculate the value of a private company using market-based methods and describe the advantages and disadvantages of...

2.

Guideline Transaction Method

Guideline Transactions Method

Prior acquisition values for entire companies are used. No additional adjustment for a controlling interest is necessary (as would be the case with GPCM).

Factors to consider when using multiples from past transactions

- 1. Transaction type.** A transaction may be either strategic or financial.
- 2. Contingent consideration.** Contingent consideration refers to that part of the acquisition price that is contingent on the achievement of specific company performance targets.
- 3. Type of consideration.** Some acquisitions are made with stock rather than cash. Comparing transactions of different consideration types may not be relevant.

[LOS 6.i] Calculate the value of a private company using market-based methods and describe the advantages and disadvantages of...

2.

Guideline Transaction Method

Example: Valuation with the guideline transactions method

Natalie Hoskins is valuing a private firm, Lafayette Furniture (Lafayette), for acquisition using GTM and enterprise value to EBITDA multiples. Hoskins delates the average public company multiple by 30% to account for the higher risk of Lafayette.

Other data are as follows:

Normalized EBITDA	\$18,200,000
Average EV/EBITDA multiple	7.2

Calculate the enterprise value of Lafayette using the GTM.

Answer:

The adjustment to the EV/EBITDA multiple for the higher risk of Lafayette is as follows:

$$7.2 \times (1 - 0.30) = 5.0$$

This adjusted multiple is applied against the normalized EBITDA to estimate the value of the firm:

$$5.0 \times \$18,200,000 = \$91,000,000$$